

Algebra I

Lecture notes

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Tablet reading edition

The material borrows heavily from Serge Lang's *Algebra*.

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1 Motivation: solving equations

Algebra starts with polynomial equations.

1. $ax + b = 0$, with $a \neq 0$, has solution $x = -b/a$.

2. $ax^2 + bx + c = 0$, with $a \neq 0$, has solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

3. For $x^3 + px + q = 0$, Cardano's formula gives

$$x = C - \frac{p}{3C}, \quad C^3 = -\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}},$$

provided $C \neq 0$ and the square and cube roots are chosen compatibly. The degenerate cases are obtained separately (or by taking a limiting form of the formula).

4. Quartic polynomials are solvable by radicals: using arithmetic operations and extraction of roots.

Theorem (Abel–Ruffini, 1824). *The general quintic is not solvable by radicals.*

Theorem (Galois, 1830). *Let K be a field with $\text{char } K = 0$, and let $f \in K[x]$. Then $f(x) = 0$ is solvable by radicals if and only if the Galois group $\text{Gal}(K_f/K)$ of the splitting field K_f of f over K is solvable.*

Example. For $f(x) = x^5 - 1 \in \mathbb{Q}[x]$, $\text{Gal}(K_f/\mathbb{Q}) \simeq (\mathbb{Z}/5\mathbb{Z})^\times$ is solvable. For $f(x) = x^5 - x - 1 \in \mathbb{Q}[x]$, $\text{Gal}(K_f/\mathbb{Q}) \simeq S_5$ is not solvable.

Course outline: I. Groups (Lang, Chapter I); II. Rings and polynomials (Lang, Chapters II–IV); III. Fields and Galois theory (Lang, Chapters V–VI).

2 Categories

Definition. A category $\mathcal{C}$ consists of a class $\text{Ob}(\mathcal{C})$ of objects, sets $\text{Hom}_{\mathcal{C}}(X, Y)$ of morphisms for each pair of objects, and composition maps

$$\begin{aligned} \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) &\longrightarrow \text{Hom}_{\mathcal{C}}(X, Z), \\ (g, f) &\longmapsto g \circ f. \end{aligned}$$

Composition is associative whenever defined. For every X there is an identity $\text{id}_X \in \text{Hom}_{\mathcal{C}}(X, X)$ satisfying $\text{id}_X \circ f = f$ and $g \circ \text{id}_X = g$ whenever the compositions are defined. Such an identity is unique.

Write $f : X \rightarrow Y$ for $f \in \text{Hom}_{\mathcal{C}}(X, Y)$. It is an *isomorphism*, written $f : X \xrightarrow{\sim} Y$, if some $g : Y \rightarrow X$ satisfies $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. We write $X \simeq Y$ if such an isomorphism exists. We also write $X \in \mathcal{C}$ and $(f : X \rightarrow Y) \in \mathcal{C}$.

Example. • **Set:** sets and set maps.

- **Vect_k:** vector spaces over a field k and k -linear maps.
- **Top:** topological spaces and continuous maps.
- **Op(X)** for $X \in \mathbf{Top}$: open subsets of X and inclusions.
- A partially ordered set $(P, \leq)$ defines a category with objects P and $\text{Hom}(x, y) = \{i_{xy}\}$ if $x \leq y$, and $\emptyset$ otherwise.

3 Monoids and groups

3.1 Monoids

Definition. A monoid $(M, *, e)$ is a set with a binary operation $*$: $M \times M \rightarrow M$ satisfying $x * (y * z) = (x * y) * z$ and $x * e = e * x = x$ for all $x, y, z \in M$. A monoid homomorphism $\varphi : M \rightarrow N$ satisfies

$$\varphi(x *_M y) = \varphi(x) *_N \varphi(y), \quad \varphi(e_M) = e_N.$$

Monoids and their homomorphisms form the category **Mon**.

The identity is unique: if e' is another identity, then $e' = e' * e = e$. Multiplicative notation is $(M, \cdot, 1)$. Elements x, y commute if $xy = yx$. A monoid is commutative if every pair commutes; additive notation is $(M, +, 0)$.

Example. 1. $(\mathbb{N}, \times, 1)$ and $(\mathbb{N}, +, 0)$ are commutative monoids.

2. For $X \in \mathcal{C}$, $\text{End}_{\mathcal{C}}(X) = \text{Hom}_{\mathcal{C}}(X, X)$ is a monoid under composition, with identity id_X .

3. If $x \in M$, let $x^n = x \cdots x$ (n factors), with $x^0 = 1$. Then $n \mapsto x^n$ is a homomorphism $(\mathbb{N}, +) \rightarrow M$.

4. The free monoid on a set X is $S(X)$, the set of finite sequences in X , with concatenation

$$(x_1 \cdots x_m)(y_1 \cdots y_n) = x_1 \cdots x_m y_1 \cdots y_n.$$

The empty sequence is its identity, and $\delta : X \hookrightarrow S(X)$ sends x to the one-letter sequence (x) .

3.2 Groups

Definition. For $x \in M \in \mathbf{Mon}$, an inverse is $y \in M$ with $xy = yx = e_M$. An element is invertible if it has an inverse; M is a group if every element is invertible. A group homomorphism is a monoid homomorphism between groups. These form the category $\mathbf{Grp}$.

Remark. 1. An inverse, if it exists, is unique and denoted x^{-1} . For $n \geq 1$, $(x^{-1})^n = (x^n)^{-1} =: x^{-n}$.

2. x is invertible if and only if left multiplication by x is bijective, if and only if right multiplication by x is bijective.
3. For a group homomorphism φ , $\varphi(x^{-1}) = \varphi(x)^{-1}$.

A group is *abelian* if its multiplication is commutative.

Example. $(\mathbb{Z}, +, 0)$ is abelian. The units $M^\times = \{x \in M : x \text{ invertible}\}$ form a group. Thus $\text{Aut}_{\mathcal{C}}(X) = \text{End}_{\mathcal{C}}(X)^\times$ is a group. The symmetric group on a set X is $S_X = \text{Aut}_{\mathbf{Set}}(X)$, and $S_n = S_{\{1, \dots, n\}}$. For $V \in \mathbf{Vect}_k$, the general linear group is $\text{GL}(V) = \text{Aut}_k(V)$; write $\text{GL}_n(k) = \text{GL}(k^n)$.

Example (Products). For $\mathcal{C} = \mathbf{Mon}$ or $\mathbf{Grp}$ and $(X_i)_{i \in I}$ in $\mathcal{C}$, the product $\prod_{i \in I} X_i$ has componentwise multiplication $(x_i)(y_i) = (x_i y_i)$ and identity (e_{X_i}) . Projections $\pi_j((x_i)) = x_j$ are morphisms. For $I = \{1, \dots, n\}$ write $X_1 \times \dots \times X_n$.

3.3 Subgroups

Definition. A subset $H \subseteq G$ is a subgroup, written $H \leq G$, if $e \in H$, $x, y \in H \Rightarrow xy \in H$, and $x \in H \Rightarrow x^{-1} \in H$. The restricted operation makes H a group, and $H \hookrightarrow G$ is a homomorphism.

Examples: $G \leq G$, the trivial subgroup $\{e\} \leq G$, intersections $\cap_i H_i \leq G$, and products $\prod_i H_i \leq \prod_i G_i$ when $H_i \leq G_i$.

Proposition. For $\varphi : G \rightarrow G'$, images of subgroups are subgroups of G' , and inverse images of subgroups are subgroups of G . In particular $\text{Im } \varphi = \varphi(G)$ and $\ker \varphi = \varphi^{-1}(\{e_{G'}\})$ are subgroups.

Definition. For $S \subseteq G$, the subgroup generated by S is

$$\langle S \rangle = \bigcap \{H \leq G : S \subseteq H\}.$$

We say S generates G if $G = \langle S \rangle$.

For subsets $S, T \subseteq G$, set $ST = \{st : s \in S, t \in T\}$, $S^n = S \cdots S$ (n factors), $S^0 = \{e\}$, $S^{-1} = \{s^{-1} : s \in S\}$, and $S^{-n} = (S^{-1})^n$. Then $S \leq G$ exactly when $e \in S$, $S^{-1} \subseteq S$, and $S^2 \subseteq S$. In general,

$$\langle S \rangle = \bigcup_{n=0}^{\infty} (S \cup S^{-1})^n.$$

For $H, K \leq G$, $HK \leq G$ if and only if $HK = KH$. The lattice of subgroups $\text{Sub}(G) = \{H : H \leq G\}$ is ordered by inclusion.

3.4 Cosets

For $H \leq G$, the left and right coset sets are $G/H = \{xH : x \in G\}$ and $H \backslash G = \{Hx : x \in G\}$. The index is $[G : H] = |G/H|$.

Lemma. $|G/H| = |H \backslash G|$.

Proof. Inversion sends xH bijectively onto $Hx^{-1} = (xH)^{-1}$, inducing the bijection $xH \mapsto Hx^{-1}$. $\square$

Proposition. If $K \leq H \leq G$, then $[G : K] = [G : H][H : K]$.

Proof. Choose representatives $S \subseteq G$ and $T \subseteq H$ with $G = \coprod_{x \in S} xH$ and $H = \coprod_{y \in T} yK$. Then $G = \coprod_{x \in S, y \in T} xyK$. Multiplication gives a bijection $S \times T \rightarrow ST$, so $[G : K] = |S||T|$. $\square$

Proposition. For a homomorphism $\varphi : G \rightarrow G'$ and $x \in G$,

$$\varphi^{-1}(\{\varphi(x)\}) = x \ker \varphi = (\ker \varphi)x.$$

Proof. Put $K = \ker \varphi$. The equality $\varphi(y) = \varphi(x)$ is equivalent to $\varphi(x^{-1}y) = e$, hence to $x^{-1}y \in K$, or $y \in xK$. The right-coset equality is similar. $\square$

Proposition. Suppose $H, K \leq G$, every element of H commutes with every element of K , $H \cap K = \{e\}$, and $HK = G$. Then $(h, k) \mapsto hk$ is an isomorphism $H \times K \xrightarrow{\sim} G$.

Proof sketch. The three assumptions give, respectively, a homomorphism, injectivity, and surjectivity. $\square$

3.5 Normal subgroups and quotient groups

Definition. A subgroup $H \leq G$ is normal, written $H \trianglelefteq G$, if $xH = Hx$, equivalently $xHx^{-1} = H$, for every $x \in G$.

Both G and $\{e\}$ are normal. A subgroup H is normal exactly when $H = \cup_{x \in G} xHx^{-1}$. Kernels of homomorphisms are normal. For $N \trianglelefteq G$, multiplication of subsets gives $(xN)(yN) = (xy)N$. This defines an associative operation on G/N with identity N and inverse $(xN)^{-1} = x^{-1}N$.

Definition. The quotient group is G/N together with $\pi_N : G \rightarrow G/N$, $x \mapsto xN$.

Proposition (Universal property). *If $\varphi : G \rightarrow G'$ satisfies $\varphi(N) = \{e\}$, there is a unique homomorphism $\varphi_N : G/N \rightarrow G'$ such that $\varphi = \varphi_N \circ \pi_N$.*

Proof sketch. Define $\varphi_N(xN) = \varphi(x)$. If $xN = yN$, their images under φ coincide, so this is well-defined. The homomorphism property, factorization, and uniqueness follow directly. $\square$

If $K \trianglelefteq G'$, then $\varphi^{-1}(K) = \ker(\pi_K \circ \varphi) \trianglelefteq G$. Let $\text{Sub}_H(G) = \{H' \leq G : H \leq H'\}$.

Theorem (Subgroup correspondence). *For $N \trianglelefteq G$, an order-preserving bijection is*

$$\text{Sub}(G/N) \longleftrightarrow \text{Sub}_N(G), \quad \widetilde{H} \longmapsto \pi_N^{-1}(\widetilde{H}), \quad H \longmapsto H/N.$$

Normal subgroups correspond to normal subgroups.

3.6 Isomorphism theorems

Theorem. For $\varphi : G \rightarrow G'$ and $K = \ker \varphi$, the map $\varphi_K : G/K \xrightarrow{\sim} \text{Im } \varphi$, $xK \mapsto \varphi(x)$, is an isomorphism.

Proof. The universal property gives φ_K with image $\text{Im } \varphi$. If $\varphi_K(xK) = e$, then $x \in K$, so $xK = K$. Thus its kernel is trivial. $\square$

Theorem. If $K \trianglelefteq H \trianglelefteq G$ and $K \trianglelefteq G$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \simeq G/H.$$

Proof. The map induced by π_H on G/K is surjective and has kernel H/K : its inverse image under π_K is $\pi_H^{-1}(\{e\}) = H$. Apply the first theorem. $\square$

Theorem. If $H \leq G$ and $K \trianglelefteq G$, then $HK \leq G$, $K \trianglelefteq HK$, $H \cap K \trianglelefteq H$, and inclusion induces

$$H/(H \cap K) \simeq HK/K.$$

Proof. The composite $H \hookrightarrow HK \rightarrow HK/K$ has image HK/K and kernel $H \cap K$. $\square$

Lemma (Zassenhaus butterfly lemma). Let $U, V \leq G$, $u \trianglelefteq U$, and $v \trianglelefteq V$. Then

$$u(U \cap v) \trianglelefteq u(U \cap V), \quad (u \cap V)v \trianglelefteq (U \cap V)v,$$

and

$$\frac{u(U \cap V)}{u(U \cap v)} \simeq \frac{(U \cap V)v}{(u \cap V)v}.$$

Proof sketch. Normality follows directly. Check the identity

$$(U \cap V) \cap u(U \cap v) = (u \cap V)(U \cap v).$$

Both quotients identify with $(U \cap V)/((u \cap V)(U \cap v))$ by the isomorphism theorem. $\square$

3.7 Towers of groups

Definition. A tower or series of subgroups of G is a finite sequence

$$G = G_0 \geq G_1 \geq \cdots \geq G_m = \{e\}.$$

It is normal if $G_{i+1} \trianglelefteq G_i$; its factors are G_i/G_{i+1} . A normal tower is abelian (respectively cyclic) if all factors are abelian (respectively cyclic). A refinement inserts finitely many subgroups.

Lang does not require the final entry of a tower to be $\{e\}$. For $\varphi : G \rightarrow G'$ and a tower (G'_i) , its pullback is the tower of inverse images $\varphi^{-1}(G'_0) \geq \cdots \geq \varphi^{-1}(G'_m) \geq \{e\}$.

Proposition. *Pullback preserves normal towers. If the tower in G' and $\ker \varphi$ are abelian (respectively cyclic), its pullback is abelian (respectively cyclic).*

Proposition. *Every abelian tower of a finite group admits a cyclic refinement.*

Proof. It suffices to show that every finite abelian group has a cyclic tower. Induct on $|G|$. The case $|G| = 1$ is clear. Otherwise choose $e \neq x \in G$. The abelian group $G/\langle x \rangle$ has smaller order, so has a cyclic tower by induction. Pull it back and append $\{e\}$. $\square$

Example. The additive group $\mathbb{Q}$ has no cyclic tower.

3.8 Solvable groups

Definition. A group is solvable if it has an abelian tower.

We will show that groups of order p^n , p prime, are solvable. The Feit–Thompson theorem states that all finite groups of odd order are solvable.

Example. In $\mathrm{GL}_n(k)$, let D be the diagonal subgroup and T the upper triangular subgroup. Let N be the strictly upper triangular matrices, and let $U = I + N$ be the subgroup with diagonal entries all 1. Set $U_t = I + N^t$. Then

$$T \supseteq U \supseteq U_2 \supseteq \cdots \supseteq U_n = \{I\}$$

is an abelian tower, so T is solvable.

Definition. A sequence $G' \xrightarrow{\varphi} G \xrightarrow{\psi} G''$ is exact at G if $\mathrm{Im} \varphi = \ker \psi$. A longer sequence is exact if it is exact at every intermediate term.

For $H \trianglelefteq G$, the sequence $1 \rightarrow H \rightarrow G \rightarrow G/H \rightarrow 1$ is exact.

Theorem. For an exact sequence $1 \rightarrow G' \xrightarrow{\varphi} G \xrightarrow{\psi} G'' \rightarrow 1$, G is solvable if and only if G' and G'' are solvable.

Proof. Pull back an abelian tower of G along φ , and take its image under ψ , to prove the forward implication. Conversely pull back an abelian tower of G'' and append an abelian tower of the kernel $\varphi(G')$. $\square$

Definition. The commutator of $x, y \in G$ is $[x, y] = xyx^{-1}y^{-1}$. The commutator or derived subgroup is $G^{(1)} = [G, G] = \langle [x, y] : x, y \in G \rangle$. Put $G^{(0)} = G$ and $G^{(n)} = (G^{(n-1)})^{(1)}$.

Exercise: G is solvable if and only if $G^{(n)} = \{e\}$ for some $n \geq 0$.

3.9 Composition series

Two normal towers $(G_i)_{i=0}^m$ and $(H_i)_{i=0}^n$ are equivalent if $m = n$ and a permutation $\sigma \in S_m$ matches their factors:

$$G_{i-1}/G_i \simeq H_{\sigma(i)-1}/H_{\sigma(i)}.$$

Theorem (Schreier). *Any two normal towers of a group have equivalent refinements.*

Proof. Define $G_{ij} = G_{i+1}(H_j \cap G_i)$ for $0 \leq i < m$, $0 \leq j \leq n$. Since $G_{i0} = G_i$ and $G_{in} = G_{i+1}$, these give a refinement with mn factors. Similarly put $H_{ji} = H_{j+1}(G_i \cap H_j)$. The butterfly lemma gives normality and

$$G_{ij}/G_{i,j+1} \simeq H_{ji}/H_{j,i+1}.$$

Thus the refinements are equivalent. □

Definition. A nontrivial group is simple if its only normal subgroups are $\{e\}$ and itself. A composition series is a normal tower with simple factors, called Jordan factors.

Every finite group has a composition series.

Theorem (Jordan–Hölder). *Any two composition series of a group are equivalent.*

Proof. By Schreier, they have equivalent refinements. If $G_i \geq H \geq G_{i+1}$ is an inserted normal step, simplicity of G_i/G_{i+1} forces $H = G_i$ or G_{i+1} . Thus refinements only repeat terms, introducing trivial factors, and the original simple factors match. □

Example. If $N = p_1^{a_1} \cdots p_n^{a_n}$, the Jordan factors of $\mathbb{Z}/N\mathbb{Z}$ are a_i copies of $\mathbb{Z}/p_i\mathbb{Z}$ for $1 \leq i \leq n$.

4 Group actions

4.1 Actions and stabilizers

Definition. A left G -set is a set X with an action $G \times X \rightarrow X$, $(g, x) \mapsto gx$, such that $g(hx) = (gh)x$ and $1x = x$. A morphism of G -sets is a map $\varphi : X \rightarrow Y$ with $\varphi(gx) = g\varphi(x)$. These form the category $G\text{-Set}$.

Actions on X correspond bijectively to homomorphisms $\rho : G \rightarrow \text{Aut}_{\text{Set}}(X)$ by $\rho(g)(x) = gx$.

Definition. The orbit of x is $O_G(x) = Gx = \{gx : g \in G\}$. Its stabilizer is $G_x = \{g \in G : gx = x\} \leq G$. A point is fixed if $Gx = \{x\}$; write $X^G = \text{Fix}_G(X)$ for all fixed points.

Example. 1. Each orbit is itself a G -set.

2. A homomorphism $G \rightarrow \text{GL}(V)$, for a k -vector space V , is a representation.
3. For $H \leq G$, left translation acts on G/H by $(g, xH) \mapsto gxH$.
4. Conjugation acts on G by ${}^gx = gxg^{-1}$, and on $\text{Sub}(G)$ by ${}^gH = gHg^{-1}$.

4.2 Orbits and the class equation

For any action, $G_{gx} = gG_xg^{-1}$.

Theorem (Orbit-stabilizer). *The map $\Phi_x : G/G_x \xrightarrow{\sim} Gx$, $gG_x \mapsto gx$, is an isomorphism of G -sets. In particular $|Gx| = [G : G_x]$.*

Proof. $gG_x = hG_x$ if and only if $h^{-1}g \in G_x$, if and only if $gx = hx$. This proves well-definedness and bijectivity; equivariance is immediate. $\square$

If S is a complete set of orbit representatives, then

$$X = \coprod_{x \in S} Gx \simeq \coprod_{x \in S} G/G_x.$$

For finite X , $|X| = \sum_{x \in S} [G : G_x]$. Under conjugation on G , x, y are conjugate if $x = gyg^{-1}$ for some g . The conjugacy class of x is ${}^Gx = \{gxg^{-1} : g \in G\}$. Its stabilizer is the centralizer $G_x = \{g : gx = xg\}$. The center is $Z(G) = \{x : gx = xg \text{ for all } g \in G\}$.

Theorem (Class formula). *For finite G ,*

$$|G| = |Z(G)| + \sum_x [G : G_x],$$

where x ranges over representatives of conjugacy classes of size at least 2.

For conjugation on subgroups, H, H' are conjugate if $H = gH'g^{-1}$. The normalizer is $N_G(H) = \{g : gHg^{-1} = H\}$; always $H \trianglelefteq N_G(H)$. The fixed subgroups are precisely the normal subgroups.

4.3 Orders and exponents

Definition. The order (period) of $x \in G$ is $\text{ord}(x) = |\langle x \rangle|$. The exponent is

$$\exp(G) = \text{lcm}\{\text{ord}(x) : x \in G\} \in \mathbb{N}_{\geq 1} \cup \{\infty\}.$$

For finite G , $\text{ord}(x)$ divides $|G|$. A group of prime order is cyclic.

Lemma. For a finite abelian group G , $|G|$ divides $\exp(G)^a$ for some $a \in \mathbb{N}$.

Proof. Induct on $|G|$, the trivial case being immediate. Choose $1 \neq x \in G$. The exponent of $G/\langle x \rangle$ divides $\exp(G)$, so induction gives $|G/\langle x \rangle| \mid \exp(G)^{a'}$. Also $\text{ord}(x) \mid \exp(G)$. Multiplication gives the result with $a = a' + 1$. $\square$

Corollary. If G is finite abelian and $p \mid |G|$ is prime, G contains an element of order p .

Proof. The lemma implies $p \mid \exp(G)$, so some x has order ps . Then x^s has order p . $\square$

Definition. A p -group has order p^n for some $n \geq 0$. A p -subgroup is a subgroup which is a p -group. A Sylow p -subgroup of finite G has order p^n where $p^n \parallel |G|$. Write $\text{Syl}_p(G)$ for the set of these subgroups.

4.4 Sylow subgroups

Theorem. *Every finite group has a Sylow p -subgroup.*

Proof. Induct on $|G|$. If a proper subgroup H has index prime to p , any Sylow p -subgroup of H , existing by induction, is one of G . Otherwise every proper subgroup has index divisible by p . The class formula implies $p \mid |Z(G)|$ (unless G is trivial, the base case). The preceding corollary gives $x \in Z(G)$ of order p . Put $H = \langle x \rangle \trianglelefteq G$. If $p^n \parallel |G|$, induction gives $P' \leq G/H$ of order p^{n-1} . Its inverse image has order p^n . $\square$

Lemma. *If a p -group H acts on finite S , then $|S^H| \equiv |S| \pmod{p}$.*

Proof. Every non-singleton orbit has size $[H : H_x]$ divisible by p . $\square$

Theorem. *For a finite group G :*

1. *Every p -subgroup is contained in a Sylow p -subgroup.*
2. *All Sylow p -subgroups are conjugate.*
3. $|\text{Syl}_p(G)| \equiv 1 \pmod{p}$.

Proof. Fix $P \in \text{Syl}_p(G)$ and let S be its conjugacy orbit. Then $|S| = [G : N_G(P)]$ divides $[G : P]$, so is prime to p . A p -subgroup H acts on S , and $|S^H| \equiv |S| \not\equiv 0 \pmod{p}$. Thus some $Q \in S$ is normalized by H . Then HQ is a subgroup and

$$|HQ| = [H : H \cap Q]|Q| \mid |H||Q|,$$

so it is a p -group. Maximality of $|Q|$ gives $HQ = Q$ and $H \leq Q$. If H is Sylow, $H = Q$, proving conjugacy. Finally the only element of S fixed by P is P , so $|S| \equiv |S^P| = 1 \pmod{p}$. $\square$

Exercise (Cauchy): if $p \mid |G|$, there exists $x \in G$ with $\text{ord}(x) = p$.

4.5 Groups with few prime factors

Theorem. *If G is a finite p -group, then G is solvable; if $|G| > 1$, then $|Z(G)| > 1$.*

Proof. The second claim follows from the class formula: every noncentral conjugacy-class size divides $|G|$ and is greater than 1, hence divisible by p , so $p \mid |Z(G)|$. For the first, use induction on $|G|$ and quotient by a central subgroup of order p . $\square$

Corollary. *A nontrivial p -group has a cyclic tower whose factors all have order p .*

Lemma. *Let p be the smallest prime dividing $|G|$. If $H \leq G$ and $[G : H] = p$, then $H \trianglelefteq G$.*

Proof. If $N_G(H) \neq G$, then $N_G(H) = H$. Conjugation on the orbit of H gives $\rho : G \rightarrow S_p$. Its kernel is contained in H . Hence

$$p[H : \ker \rho] = [G : \ker \rho] \mid p!,$$

so some prime $q < p$ divides $[H : \ker \rho]$ and $|G|$, a contradiction. $\square$

Proposition. *A group of order pq , for distinct primes p, q , is solvable.*

Proof. Suppose $p < q$ and choose $K \in \text{Syl}_q(G)$. Then K is cyclic and has index p , hence is normal. The quotient G/K is cyclic. $\square$

5 Symmetric and alternating groups

5.1 Cycles

Let $J_n = \{1, \dots, n\}$ and $S_n = \text{Aut}_{\text{Set}}(J_n)$, composed using the two-line notation

$$\sigma = \begin{pmatrix} 1 & \cdots & n \\ \sigma(1) & \cdots & \sigma(n) \end{pmatrix}.$$

A cycle of σ is an orbit of $\langle \sigma \rangle$ in J_n . A k -cycle has one orbit $(a_1 \cdots a_k)$ of size k , with $\sigma(a_i) = a_{i+1}$ and $\sigma(a_k) = a_1$, and fixes all other points. A transposition is a 2-cycle. Disjoint cycles have disjoint supports and commute. A cycle decomposition is $\sigma = \sigma_1 \cdots \sigma_m$ into pairwise disjoint cycles whose supports cover J_n . Its cycle type is the multiset of cycle lengths.

Proposition. 1. S_n is generated by transpositions.

2. Every permutation has a cycle decomposition, unique up to reordering.

3. Two elements of S_n are conjugate exactly when they have the same cycle type.

Proof. Induct on n for (1), using the stabilizer H of n , isomorphic to S_{n-1} , and the coset decomposition $S_n = H \sqcup \coprod_{k=1}^{n-1} (kn)H$. For (2), take the orbits. For (3), if σ and σ' have corresponding cycles, define γ to carry every entry in each cycle of σ to the corresponding entry of σ' ; then $\gamma\sigma\gamma^{-1} = \sigma'$. $\square$

For example,

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 6 & 4 & 1 & 2 & 5 & 7 \end{pmatrix} = (1\ 3\ 4)(2\ 6\ 5)(7).$$

5.2 Sign of a permutation

The action of S_n on $\mathcal{F}_n = \text{Hom}_{\text{Set}}(\mathbb{Z}^n, \mathbb{Z})$ is

$$(\sigma f)(x_1, \dots, x_n) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

It respects addition and multiplication.

Proposition. *There is a unique homomorphism $\text{sgn} : S_n \rightarrow \{\pm 1\}$ taking every transposition to -1 .*

Proof. Consider the Vandermonde function

$$\Delta(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

A transposition changes its sign, so define $\text{sgn}(\sigma)$ by $\sigma\Delta = \text{sgn}(\sigma)\Delta$. Uniqueness follows because transpositions generate S_n . $\square$

Thus a product of m transpositions has sign $(-1)^m$. A permutation is even or odd according as its sign is 1 or -1 . The alternating group is $A_n = \ker(\text{sgn})$.

5.3 The alternating group

Lemma. A_n is generated by 3-cycles.

Proof. Pair transpositions in an expression for an even permutation. If (ij) and (rs) share a letter, their product is a 3-cycle or the identity. If their supports are disjoint, $(ij)(rs) = (ijr)(jrs)$. $\square$

Lemma. For $n \geq 5$, all 3-cycles are conjugate in A_n .

Proof. A permutation carrying the three entries of one cycle to the other conjugates them in S_n . If it is odd, compose it with a transposition supported away from those entries; this does not change the conjugation and makes it even. $\square$

Theorem. For $n \geq 5$, A_n is simple.

Proof. Let $1 \neq N \trianglelefteq A_n$. It suffices to find a 3-cycle in N . Choose nonidentity $\sigma \in N$ moving as few letters as possible. If all its nontrivial cycles are transpositions, there are at least two, say (ij) and (rs) . Choose k outside these four letters and put $\tau = (rsk)$. The commutator $\sigma' = \tau\sigma\tau^{-1}\sigma^{-1} \in N$ is nonidentity, fixes i, j , and fixes every original fixed point except possibly k . It therefore moves fewer letters, a contradiction. Otherwise write a cycle beginning $(ijk \dots)$. If σ is not the single 3-cycle (ijk) , parity implies that it moves at least two more letters r, s . Put $\tau = (rsk)$. The supports of τ and $\sigma\tau\sigma^{-1}$ differ: equality would make the three-element set $\{r, s, k\}$ invariant under σ , though the cycle of k also contains i, j . Thus $\sigma' = \tau\sigma\tau^{-1}\sigma^{-1}$ is nonidentity. It fixes every original fixed point, because r, s, k are moved by σ , and also fixes j , since $\sigma^{-1}(j) = i$. Again its support is smaller, a contradiction. Hence σ is a 3-cycle. Conjugacy and generation give $N = A_n$. $\square$

Corollary. For $n \geq 5$, S_n is not solvable.

6 Abelian groups

6.1 Direct sums

Let $\mathbf{Ab}$ be the category of abelian groups. For a family $(A_i)_{i \in I}$, define

$$\bigoplus_{i \in I} A_i = \{(a_i) \in \prod_{i \in I} A_i : a_i = 0 \text{ for almost all } i\}.$$

The canonical injection $\lambda_j : A_j \rightarrow \bigoplus_i A_i$ has j th component a and all others 0. For finite I , direct sum and direct product coincide. If all $A_i = A$, write $A^{(I)} = \bigoplus_I A$ and $A^I = \prod_I A$; these are respectively finitely supported and arbitrary functions $I \rightarrow A$.

Proposition (Universal property). *Given $f_i : A_i \rightarrow B$ in $\mathbf{Ab}$, there is a unique $f : \bigoplus_i A_i \rightarrow B$ with $f_i = f \circ \lambda_i$, namely*

$$f((x_i)) = \sum_i f_i(x_i),$$

where the sum is finite.

If $A_i \leq A$, summation gives $\bigoplus_i A_i \rightarrow A$. When this is an isomorphism we write $A = \bigoplus_i A_i$. In particular, $A = A_1 \oplus A_2$ exactly when $A = A_1 + A_2$ and $A_1 \cap A_2 = 0$.

6.2 Free abelian groups and bases

Definition. The free abelian group on a set S is $\mathbb{Z}^{(S)}$, the finitely supported functions $S \rightarrow \mathbb{Z}$. The map $\delta : S \rightarrow \mathbb{Z}^{(S)}$ sends x to δ_x , with value 1 at x and 0 elsewhere. Often write x for δ_x , so an element is a finite sum $\sum_{x \in S} a_x x$.

Proposition (Universal property). *A set map $f : S \rightarrow B$, $B \in \mathbf{Ab}$, extends uniquely to a homomorphism $\varphi_f : \mathbb{Z}^{(S)} \rightarrow B$, given by*

$$\varphi_f \left(\sum_x a_x x \right) = \sum_x a_x f(x).$$

A map $f : S \rightarrow S'$ therefore induces a unique $\bar{f} : \mathbb{Z}^{(S)} \rightarrow \mathbb{Z}^{(S')}$ with $\bar{f} \circ \delta_S = \delta_{S'} \circ f$.

$$\begin{array}{ccc} S & \xrightarrow{\delta_S} & \mathbb{Z}^{(S)} \\ f \downarrow & & \downarrow \bar{f} \\ S' & \xrightarrow{\delta_{S'}} & \mathbb{Z}^{(S')} \end{array}$$

Definition. An abelian group A is free if $A \simeq \mathbb{Z}^{(S)}$ for some set S . A subset $S \subseteq A$ is a basis if the map induced by inclusion is an isomorphism. Equivalently, every element has a unique expression as a finite integral linear combination of elements of S .

Proposition. *Any two bases of a free abelian group have the same cardinality, called its rank.*

Proof. If $\mathbb{Z}^{(S)} \simeq \mathbb{Z}^{(T)}$, reduction modulo 2 gives $(\mathbb{Z}/2\mathbb{Z})^{(S)} \simeq (\mathbb{Z}/2\mathbb{Z})^{(T)}$. If infinite, the underlying set has cardinality $|S|$ or $|T|$; if finite, its size is $2^{|S|}$ or $2^{|T|}$. Hence $|S| = |T|$. $\square$

6.3 Subgroups of free abelian groups

For free groups A_i , the images of their bases form a basis of $\bigoplus_i A_i$, so its rank is $\sum_i \text{rank}(A_i)$.

Lemma (Splitting). *If $0 \rightarrow \ker \pi \rightarrow A \xrightarrow{\pi} F \rightarrow 0$ is exact and F is free, there is $s : F \rightarrow A$ with $\pi s = \text{id}_F$, and $A = \ker \pi \oplus \text{Im } s$.*

Proof. Choose a lift in A for each basis vector of F , and extend linearly to s . Then $\pi s = \text{id}_F$. For $a \in A$, $a - s\pi(a) \in \ker \pi$, so the two subgroups sum to A . If $s(b) \in \ker \pi$, then $b = \pi s(b) = 0$, proving trivial intersection. $\square$

Thus ranks add in a short exact sequence of free abelian groups.

Theorem. *A subgroup B of a free abelian group A is free and has rank at most that of A .*

Proof. Well-order a basis $(x_\alpha)_{\alpha < \kappa}$ of A , and put $A_\alpha = \bigoplus_{\beta < \alpha} \mathbb{Z}x_\beta$ and $B_\alpha = B \cap A_\alpha$. At a successor stage, $B_{\alpha+1}/B_\alpha$ embeds in $A_{\alpha+1}/A_\alpha \simeq \mathbb{Z}$, so it is either zero or infinite cyclic. The resulting short exact sequence splits. Choose a lift y_α whenever this quotient is nonzero. At a limit ordinal take unions. Transfinite induction then gives

$$B_\alpha = \bigoplus_{\beta < \alpha, y_\beta \text{ chosen}} \mathbb{Z}y_\beta.$$

At $\alpha = \kappa$ this is a basis of B , with at most one basis element for each basis element of A . $\square$

6.4 Linear algebra over the integers

In the standard bases, a homomorphism $\varphi : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$ corresponds to the integral matrix M_φ whose j th column is $\varphi(e_j)$. Then $\varphi(x) = M_\varphi x$, $M_{\text{id}} = I$, and $M_{\psi \circ \varphi} = M_\psi M_\varphi$. Thus $\text{Aut}(\mathbb{Z}^n) \simeq \text{GL}_n(\mathbb{Z})$, the matrices invertible over $\mathbb{Z}$.

Proposition (Smith normal form). *For $M \in \text{Mat}_{m \times n}(\mathbb{Z})$, there are $B \in \text{GL}_m(\mathbb{Z})$ and $C \in \text{GL}_n(\mathbb{Z})$ such that $D = BMC$ is diagonal and $D_{11} \mid D_{22} \mid \cdots \mid D_{NN}$, $N = \min(m, n)$.*

Basic case; general proof in Problem Set 4. For a nonzero column $M = (a_1, a_2)^t$, put $d = \gcd(a_1, a_2)$ and choose $b_1 a_1 + b_2 a_2 = d$. Then

$$\begin{pmatrix} b_1 & b_2 \\ -a_2/d & a_1/d \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} d \\ 0 \end{pmatrix},$$

and the multiplying matrix has integral inverse

$$\begin{pmatrix} a_1/d & -b_2 \\ a_2/d & b_1 \end{pmatrix}.$$

□

Corollary. *Every finitely generated abelian group is a product of finitely many cyclic groups.*

Proof. Choose an exact presentation $0 \rightarrow \mathbb{Z}^n \xrightarrow{M} \mathbb{Z}^m \rightarrow A \rightarrow 0$. Apply Smith normal form to get the same cokernel as a diagonal matrix D .

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z}^n & \xrightarrow{M} & \mathbb{Z}^m & \longrightarrow & A \longrightarrow 0 \\ & & \downarrow C^{-1} & & \downarrow B & & \parallel \\ 0 & \longrightarrow & \mathbb{Z}^n & \xrightarrow{D} & \mathbb{Z}^m & \longrightarrow & A \longrightarrow 0 \end{array}$$

Then $A \simeq \bigoplus_{k=1}^m \mathbb{Z}/d_k \mathbb{Z}$, with $d_k = D_{kk}$ when a diagonal entry exists and $d_k = 0$ otherwise. □

6.5 Torsion abelian groups

An element is torsion if it has finite order. The torsion subgroup is $A_{\text{tor}} = \{x \in A : \text{ord}(x) < \infty\}$; A is torsion if $A = A_{\text{tor}}$. Set $A[m] = \ker(m : A \rightarrow A)$ and, for prime p , $A(p) = \bigcup_{k \geq 0} A[p^k]$.

Lemma. *For positive integers m, n ,*

$$A[m] \cap A[n] = A[\gcd(m, n)], \quad A[m] + A[n] = A[\text{lcm}(m, n)].$$

In particular, if $\gcd(m, n) = 1$, then $A[mn] = A[m] \oplus A[n]$.

Proof. Write $d = \gcd(m, n) = mr + ns$ with $r, s \in \mathbb{Z}$. An element killed by both m, n is killed by d . If $(mn/d)x = 0$, then

$$x = (mr/d)x + (ns/d)x,$$

where the first summand is killed by n and the second by m . The reverse inclusions are immediate. $\square$

A homomorphism $\varphi : A \rightarrow B$ restricts to $\varphi_m : A[m] \rightarrow B[m]$. For torsion groups, φ is injective (respectively an isomorphism) if and only if every φ_m is. Also $(\bigoplus_i A_i)[m] = \bigoplus_i A_i[m]$.

Theorem. $A_{\text{tor}} = \bigoplus_{p \text{ prime}} A(p)$.

Proof. Summation defines a map from the direct sum to A_{tor} . On elements killed by m , the preceding lemma gives the isomorphism

$$\bigoplus_{p^r \parallel m} A[p^r] \xrightarrow{\sim} A[m].$$

The restriction criterion proves the assertion. $\square$

6.6 Structure theorem

Theorem. *A finite abelian p -group has a unique expression up to isomorphism*

$$A \simeq \bigoplus_{i=1}^s \mathbb{Z}/p^{r_i}\mathbb{Z}, \quad r_1 \geq \cdots \geq r_s \geq 1.$$

Proof. Existence follows from the preceding results. If two such decompositions exist, apply induction to pA . This determines all exponents at least 2, after decreasing them by 1. Comparing the orders then determines the number of exponents equal to 1. $\square$

Theorem. *A finitely generated torsion-free abelian group is free of finite rank.*

Proof. Choose a maximal $\mathbb{Z}$ -linearly independent set $x_1, \dots, x_n$ and let $B = \langle x_1, \dots, x_n \rangle$. If $y_1, \dots, y_m$ generate A , maximality gives nonzero integers b_j such that $b_j y_j \in B$. Put $b = b_1 \cdots b_m$. Multiplication by b identifies A with $bA \leq B$. Since subgroups of free abelian groups are free, so is A . $\square$

Theorem. *For a finitely generated abelian group,*

$$A \simeq A_f \oplus A_{\text{tor}}, \quad A_{\text{tor}} \simeq \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_r\mathbb{Z},$$

where A_f is free of finite rank and $2 \leq d_1 \mid \cdots \mid d_r$ are unique.

Proof. The quotient A/A_{tor} is finitely generated and torsion-free, hence free. The sequence $0 \rightarrow A_{\text{tor}} \rightarrow A \rightarrow A/A_{\text{tor}} \rightarrow 0$ splits. The finite part is described by the preceding primary decompositions. $\square$

6.7 Homomorphisms and exactness

For abelian groups A, B , $\text{Hom}(A, B)$ is abelian under pointwise operations. A map $\varphi : A_1 \rightarrow A_2$ gives pullback $\varphi^* : \text{Hom}(A_2, B) \rightarrow \text{Hom}(A_1, B)$, $f \mapsto f\varphi$. A map $\psi : B_1 \rightarrow B_2$ gives pushforward $\psi_* : \text{Hom}(A, B_1) \rightarrow \text{Hom}(A, B_2)$, $f \mapsto \psi f$. These preserve identities and satisfy $(\psi\varphi)^* = \varphi^*\psi^*$ and $(\psi\varphi)_* = \psi_*\varphi_*$. Recall: $0 \rightarrow A' \xrightarrow{\varphi} A$ is exact exactly when φ is injective; $A \xrightarrow{\psi} A'' \rightarrow 0$ is exact exactly when ψ is surjective.

Proposition. *If $0 \rightarrow A' \xrightarrow{\varphi} A \xrightarrow{\psi} A'' \rightarrow 0$ is exact, then both sequences*

$$\begin{aligned} 0 \rightarrow \text{Hom}(A'', B) &\xrightarrow{\psi^*} \text{Hom}(A, B) \xrightarrow{\varphi^*} \text{Hom}(A', B), \\ 0 \rightarrow \text{Hom}(B, A') &\xrightarrow{\varphi_*} \text{Hom}(B, A) \xrightarrow{\psi_*} \text{Hom}(B, A'') \end{aligned}$$

are exact.

Proof. Surjectivity of ψ makes ψ^* injective. If $f\varphi = 0$, f factors uniquely through $A/\text{Im } \varphi \simeq A''$, proving exactness of the first sequence.

$$\begin{array}{ccc} A & \xrightarrow{\psi} & A'' \\ & \searrow f & \downarrow \exists! \bar{f} \\ & & B \end{array}$$

Injectivity of φ makes φ_* injective. If $\psi f = 0$, then $\text{Im } f \subseteq \ker \psi = \text{Im } \varphi$, so f factors through A' . $\square$

6.8 Dual groups

The torsion group $\mathbb{Q}/\mathbb{Z}$ has a unique subgroup of order m , namely $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$, which is cyclic. Fix $m \geq 1$. Let $\mathbf{Ab}_m$ consist of groups killed by m . For $A \in \mathbf{Ab}_m$, define

$$\widehat{A} = \text{Hom}(A, \frac{1}{m}\mathbb{Z}/\mathbb{Z}).$$

A map $\varphi : A \rightarrow B$ induces $\widehat{\varphi} : \widehat{B} \rightarrow \widehat{A}$ by $f \mapsto f\varphi$. Thus $\widehat{\widehat{\text{id}}} = \text{id}$ and $\widehat{\psi\varphi} = \widehat{\varphi}\widehat{\psi}$. If $m \mid n$, inclusion of $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$ in $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$ induces an isomorphism on $\text{Hom}(A, -)$ for $A \in \mathbf{Ab}_m$. Consequently the definition does not depend on the chosen annihilating integer; we may write $\widehat{A} = \text{Hom}(A, \mathbb{Q}/\mathbb{Z})$.

Theorem. *For finite abelian groups, there is a natural isomorphism $\widehat{A_1 \times A_2} \simeq \widehat{A_1} \times \widehat{A_2}$. Every finite abelian group A is noncanonically isomorphic to $\widehat{\widehat{A}}$, while evaluation gives a natural isomorphism $A \xrightarrow{\sim} \widehat{\widehat{A}}$.*

Proof. With projections π_i and inclusions λ_i , the first isomorphism sends (φ_1, φ_2) to $\varphi_1\pi_1 + \varphi_2\pi_2$; its inverse sends φ to $(\varphi\lambda_1, \varphi\lambda_2)$. They are inverse because $\pi_i\lambda_j = \delta_{ij}$ and $\lambda_1\pi_1 + \lambda_2\pi_2 = \text{id}$. By decomposition into cyclic groups, the second assertion reduces to $\text{Hom}(\mathbb{Z}/n\mathbb{Z}, \frac{1}{n}\mathbb{Z}/\mathbb{Z}) \simeq \frac{1}{n}\mathbb{Z}/\mathbb{Z} \simeq \mathbb{Z}/n\mathbb{Z}$. $\square$

6.9 Bilinear maps

Definition. A pairing $B : A \times A' \rightarrow C$ of abelian groups is bilinear if it is a homomorphism in each variable separately. We also write $\langle \ , \ \rangle$.

Pairings correspond to maps $L_B : A \rightarrow \text{Hom}(A', C)$, $x \mapsto B(x, -)$, or $R_B : A' \rightarrow \text{Hom}(A, C)$, $x' \mapsto B(-, x')$. Evaluation $A \times \text{Hom}(A, C) \rightarrow C$ is an example. Subsets $S \subseteq A$ and $S' \subseteq A'$ are orthogonal if $B(S, S') = 0$. Define the left and right kernels

$$L = \ker L_B = \{x : B(x, A') = 0\}, \quad R = \ker R_B = \{x' : B(A, x') = 0\}.$$

The pairing induces injections $\Phi : A/L \hookrightarrow \text{Hom}(A'/R, C)$ and $\Psi : A'/R \hookrightarrow \text{Hom}(A/L, C)$.

Theorem. If $C = \frac{1}{m}\mathbb{Z}/\mathbb{Z}$ and A/L is finite, then A'/R is finite and $\Phi : A/L \xrightarrow{\sim} \widehat{A'/R}$.

Proof. The second injection shows A'/R is finite and killed by m . The first then shows the same torsion property for A/L . Finite duality gives $|A/L| \leq |A'/R|$ and, symmetrically, $|A'/R| \leq |A/L|$. Thus the injection Φ is an isomorphism. $\square$

7 Limits and completions

7.1 Inverse limits

A partially ordered set has a reflexive, antisymmetric, transitive relation $\leq$. Its opposite reverses this relation. An I -indexed system in a category has objects G_i and compatible transition maps $f_{ij} : G_i \rightarrow G_j$ for $i \leq j$. An inverse system reverses these arrows: $f_{ij} : G_j \rightarrow G_i$, satisfying $f_{ij}f_{jk} = f_{ik}$ and $f_{ii} = \text{id}$. For $\mathcal{C} \in \{\text{Set}, \text{Mon}, \text{Grp}, \text{Ab}\}$, define

$$\varprojlim_{i \in I} G_i = \{(x_i) \in \prod_i G_i : f_{ij}(x_j) = x_i \text{ if } i \leq j\},$$

with coordinate projections π_j .

Proposition (Universal property). *Compatible maps $\varphi_i : H \rightarrow G_i$, with $f_{ij}\varphi_j = \varphi_i$, induce a unique $\varphi : H \rightarrow \varprojlim_i G_i$ such that $\pi_i\varphi = \varphi_i$.*

Proof. Take $\varphi(h) = (\varphi_i(h))_i$. □

Equivalently, $\text{Hom}(H, \varprojlim_i G_i) \simeq \varprojlim_i \text{Hom}(H, G_i)$, with transition maps given by postcomposition with f_{ij} . A poset is directed if any two elements have a common upper bound. A directed inverse system has a directed indexing poset. In Lang, “inverse limit” means directed inverse limit.

7.2 Examples and completions

For prime p , the natural reduction maps $\mathbb{Z}/p^n\mathbb{Z} \rightarrow \mathbb{Z}/p^m\mathbb{Z}$ ($m \leq n$) give

$$\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z},$$

the group (and ring) of p -adic integers. For a family I of normal subgroups, the natural quotient maps give $G \rightarrow \varprojlim_{H \in I} G/H$, $x \mapsto (xH)_H$. Taking all normal subgroups of finite index defines the profinite completion $\widehat{G} = \varprojlim_N G/N$. For a descending sequence $G = G_0 \geq G_1 \geq \cdots$ of normal subgroups, a sequence (x_k) in G is Cauchy if, for every n , there exists $N(n)$ such that $x_k^{-1}x_\ell \in G_n$ for $k, \ell \geq N(n)$. It is null if eventually $x_k \in G_n$ for every n . The completion is the group of Cauchy sequences modulo the normal subgroup of null sequences.

Theorem. *This completion is naturally isomorphic to $\varprojlim_n G/G_n$.*

Proof. A Cauchy sequence determines its eventual coset in each G/G_n . Conversely, choose representatives x_n of a compatible family of cosets. They form a Cauchy sequence, independent modulo null sequences of the representatives. These constructions are inverse. $\square$

8 Functors and universal constructions

8.1 Functors

Definition. A covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns an object $F(X)$ to each X and a map $F : \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$, preserving identities and composition: $F(\text{id}_X) = \text{id}_{F(X)}$ and $F(gf) = F(g)F(f)$.

Identity functors and composites of functors are defined on objects and morphisms in the evident way.

Example. • $h^A = \text{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \rightarrow \mathbf{Set}$ sends $f : X \rightarrow Y$ to postcomposition $\varphi \mapsto f\varphi$.

- The free monoid functor $S : \mathbf{Set} \rightarrow \mathbf{Mon}$ sends a map f to $(x_1 \cdots x_n) \mapsto (f(x_1) \cdots f(x_n))$.
- Abelianization $(-)^{\text{ab}} : \mathbf{Grp} \rightarrow \mathbf{Ab}$, the free abelian group functor $\mathbb{Z}^{(-)} : \mathbf{Set} \rightarrow \mathbf{Ab}$, and m -torsion $(-)[m] : \mathbf{Ab} \rightarrow \mathbf{Ab}_m$.
- An I -indexed system is a functor $I \rightarrow \mathcal{C}$; an inverse system is a functor $I^{\text{op}} \rightarrow \mathcal{C}$.

8.2 Contravariance and natural transformations

The opposite category $\mathcal{C}^{\text{op}}$ has the same objects, with $\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) = \text{Hom}_{\mathcal{C}}(Y, X)$ and reversed composition. A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is a functor $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$. Examples are $h_A = \text{Hom}_{\mathcal{C}}(-, A)$, duality $(-)^{\wedge} : \mathbf{Ab}_m^{\text{op}} \rightarrow \mathbf{Ab}_m$, and the vector-space dual $D = \text{Hom}_k(-, k) : \mathbf{Vect}_k^{\text{op}} \rightarrow \mathbf{Vect}_k$.

Definition. A natural transformation $\theta : F \rightarrow G$ assigns $\theta_X : F(X) \rightarrow G(X)$ such that for every $f : X \rightarrow Y$,

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \theta_X \downarrow & & \downarrow \theta_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

It is a natural isomorphism if every θ_X is an isomorphism. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is an equivalence if some $G : \mathcal{D} \rightarrow \mathcal{C}$ has $FG \simeq \text{id}_{\mathcal{D}}$ and $GF \simeq \text{id}_{\mathcal{C}}$ naturally.

Natural transformations have identities and composition. For vector spaces, evaluation gives a natural map $\theta : \text{id} \rightarrow D^2$, with $\theta_V(x)(\varphi) = \varphi(x)$.

8.3 Universal properties: covariant version

An object X is initial if there is a unique morphism $X \rightarrow Y$ for every Y . Any two initial objects are uniquely isomorphic: the unique arrows between them have identity composites by uniqueness. For a functor $F : \mathcal{C} \rightarrow \mathbf{Set}$, form the category of pairs (X, a) , $a \in F(X)$, with morphisms $f : (X, a) \rightarrow (Y, b)$ satisfying $F(f)(a) = b$. The manuscript denotes this category $(\mathcal{C}, F)$ (nonstandard notation). A pair (X, a) is universal for F if it is initial in this category. Thus, for every $b \in F(Y)$, there is a unique $f : X \rightarrow Y$ with $F(f)(a) = b$.

Example. 1. (X, id_X) is universal for $\text{Hom}_{\mathcal{C}}(X, -)$.

2. For $N \trianglelefteq G$, define $F(H) = \{\varphi : G \rightarrow H : \varphi(N) = 1\}$. Postcomposition makes this a functor, and $(G/N, \pi_N)$ is universal for it.

3. $(\mathbb{Z}^{(S)}, \delta)$ is universal for $\text{Hom}_{\mathbf{Set}}(S, -) : \mathbf{Ab} \rightarrow \mathbf{Set}$.

A pair (X, a) is universal if and only if there is a natural isomorphism $\theta : \text{Hom}_{\mathcal{C}}(X, -) \xrightarrow{\sim} F$ with $a = \theta_X(\text{id}_X)$. We call (X, θ) a representation of F ; such a functor is representable.

8.4 Free groups

Definition. A free group on a set X is a group F_X with $\delta : X \rightarrow F_X$, universal for $\text{Hom}_{\text{Set}}(X, -) : \text{Grp} \rightarrow \text{Set}$.

Theorem. A free group exists on every set, is unique up to unique isomorphism compatible with δ , and δ is injective.

Lemma. There is a set-indexed family (G_i) containing, up to isomorphism, every group generated by the image of a map from X .

Proof. Such a group has cardinality at most $\max(\aleph_0, |X|)$, since it consists of finite words in the image and its inverses. Choose a set T of this cardinality and list all group structures on all subsets of T . $\square$

Proof of the theorem. Put

$$F_0 = \prod_i \text{Hom}_{\text{Set}}(\text{Hom}_{\text{Set}}(X, G_i), G_i),$$

using pointwise group operations. Define $\delta(x)_i(f) = f(x)$ and let $F_X = \langle \text{Im } \delta \rangle$. Given $f : X \rightarrow G$, replace G by $\langle f(X) \rangle$ and choose an isomorphism $\lambda : G \rightarrow G_i$. Evaluation defines

$$\Phi_f(a) = \lambda^{-1}(\pi_i(a)(\lambda f)).$$

It is a homomorphism with $\Phi_f \delta = f$. Restrict to F_X ; uniqueness follows since $\delta(X)$ generates it. Distinct points of X can be distinguished by maps to a group, proving injectivity of δ . $\square$

The universal property produces the free group functor $F : \text{Set} \rightarrow \text{Grp}$: a map $f : X \rightarrow X'$ induces the unique homomorphism $F_X \rightarrow F_{X'}$ extending $\delta_{X'} f$.

8.5 Presentations and coproducts

A presentation of G consists of generators S and relations $R \subseteq F_S$ such that $G \simeq F_S / \langle gRg^{-1} : g \in F_S \rangle$. Write $G = \langle S \mid R \rangle$.

Example. $\langle a \mid a^n \rangle$ is cyclic of order n ; $\langle x, y \mid [x, y] \rangle \simeq \mathbb{Z}^2$. Exercise: $\langle x, y, z \mid [x, y]y^{-1}, [y, z]z^{-1}, [z, x]x^{-1} \rangle = 1$.

A coproduct of (X_i) is an object $\coprod_i X_i$ with maps $\lambda_i : X_i \rightarrow \coprod_i X_i$, universal for $\coprod_i \text{Hom}_{\mathcal{C}}(X_i, -)$. In **Set** this is disjoint union; in **Ab** it is direct sum; in a poset it is a least upper bound.

Proposition. *Coproducts exist in Grp and are called free products, written $*_{i \in I} G_i$.*

Proof. Choose presentations $G_i = \langle S_i \mid R_i \rangle$ with disjoint generator sets. Then $\langle \coprod_i S_i \mid \cup_i R_i \rangle$ has the required universal property. $\square$

8.6 Universal properties: contravariant version

An object X is final (terminal) if every Y has a unique map $Y \rightarrow X$; equivalently, X is initial in $\mathcal{C}^{\text{op}}$. Final objects are unique up to unique isomorphism. For $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$, a universal pair (X, a) has the property that, for every $b \in F(Y)$, there is a unique $f : Y \rightarrow X$ with $F(f)(a) = b$. In particular (X, id_X) is universal for $\text{Hom}_{\mathcal{C}}(-, X)$. A product of (X_i) is a pair $(\prod_i X_i, (\pi_i))$ universal for $\prod_i \text{Hom}_{\mathcal{C}}(-, X_i)$. Products in sets, monoids, groups, and abelian groups are the familiar ones. In a poset a product is a greatest lower bound. The limit of $X : I^{\text{op}} \rightarrow \mathcal{C}$ represents the functor $\varprojlim_i \text{Hom}_{\mathcal{C}}(-, X_i)$. The same notion extends to diagrams indexed by a category whose objects form a set.

8.7 Fiber products

For $Z \in \mathcal{C}$, the category $\mathcal{C}_Z$ has objects $f_X : X \rightarrow Z$ and commuting triangles as morphisms. The fiber product $X \times_Z Y$ is a product in $\mathcal{C}_Z$.

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & \searrow f_X & \swarrow f_Y \\ & Z & \end{array}$$

Its projections satisfy $f_X \pi_X = f_Y \pi_Y$, and compatible maps $\varphi_X : W \rightarrow X$, $\varphi_Y : W \rightarrow Y$ induce a unique $\varphi : W \rightarrow X \times_Z Y$ with $\pi_X \varphi = \varphi_X$ and $\pi_Y \varphi = \varphi_Y$. The map $X \times_Z Y \rightarrow X$ is the pullback of f_Y along f_X .

$$\begin{array}{ccccc} W & & & & \\ & \searrow \varphi_Y & & & \\ & & X \times_Z Y & \xrightarrow{\pi_Y} & Y \\ & \searrow \varphi_X & \downarrow \pi_X & & \downarrow f_Y \\ & & X & \xrightarrow{f_X} & Z \end{array}$$

(Note: In the original image, a dashed arrow from W to $X \times_Z Y$ is labeled $\exists! \varphi$.)

8.8 Examples of fiber products

Example. 1. If Z is terminal, $\mathcal{C}_Z$ is equivalent to $\mathcal{C}$ and $X \times_Z Y \simeq X \times Y$.

2. In sets, $X \times_Z Y = \{(x, y) : f_X(x) = f_Y(y)\}$. Pullback of a point is a fiber; for subsets of Z , the fiber product is intersection.

3. Fiber products in **Mon**, **Grp**, **Ab** are computed on underlying sets. For reduction maps to $\mathbb{Z}/5\mathbb{Z}$,

$$\mathbb{Z}/10\mathbb{Z} \times_{\mathbb{Z}/5\mathbb{Z}} \mathbb{Z}/15\mathbb{Z} \simeq \mathbb{Z}/30\mathbb{Z}.$$

4. The equalizer of $f_1, f_2 : X \rightarrow Z$ is the pullback of the diagonal $X \rightarrow X \times X$ and the map from $X \times_{f_1, Z, f_2} X$ to $X \times X$. In sets it is $\{x : f_1(x) = f_2(x)\}$.

$$\begin{array}{ccc} \text{eq}(f_1, f_2) & \longrightarrow & X \times_{f_1, Z, f_2} X \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta} & X \times X \end{array}$$

5. A vector bundle $E \rightarrow Y$ pulls back along $\varphi : X \rightarrow Y$ to $\varphi^*E = X \times_Y E \rightarrow X$.

8.9 Fiber coproducts

The category $\mathcal{C}^Z$ has objects $Z \rightarrow X$. Its coproduct $X \amalg_Z Y$ is called a fiber coproduct or pushout. Exercise: write its universal property.

$$\begin{array}{ccc} & Z & \\ f_X \swarrow & & \searrow f_Y \\ X & \xrightarrow{u} & Y \end{array}$$

If Z is initial this is the ordinary coproduct. In sets,

$$X \amalg_Z Y = (X \sqcup Y) / (f_X(z) \sim f_Y(z)).$$

Thus $X \cup Y \simeq X \amalg_{X \cap Y} Y$ for subsets of a set. Pushouts exist in monoids, groups, and abelian groups. In groups,

$$G_1 \amalg_G G_2 = (G_1 * G_2) / \langle\langle \varphi_1(g)\varphi_2(g)^{-1} : g \in G \rangle\rangle,$$

where double brackets mean normal closure. Let $\mathbf{Top}_*$ be pointed topological spaces and $\pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Grp}$ the fundamental-group functor.

Theorem (Van Kampen). *If $X = U_1 \cup U_2$, with U_1, U_2 open and path-connected and $x_0 \in U_1 \cap U_2$, and their intersection is path-connected, then*

$$\pi_1(X, x_0) \simeq \pi_1(U_1, x_0) \amalg_{\pi_1(U_1 \cap U_2, x_0)} \pi_1(U_2, x_0).$$

9 Rings

9.1 Definitions and categorical constructions

Definition. A ring $(A, +, \cdot, 0, 1)$ has an abelian additive group and a multiplicative monoid, with multiplication bilinear over addition (the distributive laws). A ring homomorphism preserves $0, 1$, addition, and multiplication. These form **Ring**; commutative rings form **CRing**.

For $A \in \mathbf{Ab}$, $\text{End}(A)$ is a ring under pointwise addition and composition; for example $\text{End}(\mathbb{Z}^n) \simeq \text{Mat}_n(\mathbb{Z})$. The zero ring $\{0\}$ is terminal in both categories; $\mathbb{Z}$ is initial. If $\varphi_A : \mathbb{Z} \rightarrow A$ is the unique map, the characteristic is the nonnegative integer n with $\ker \varphi_A = n\mathbb{Z}$. In particular $na = 0$ for every $a \in A$. Products are formed componentwise on underlying sets, and inverse limits and fiber products are also computed on underlying sets.

9.2 Basic notions

A subring $S \subseteq A$ contains $0, 1$ and is closed under addition, additive inverses, and multiplication. Its inclusion is a ring homomorphism. The center $Z(A) = \{x : xy = yx \text{ for all } y \in A\}$ is a subring, as is the image of any ring homomorphism. The subring generated by a subset $S \subseteq B$ is the intersection of all subrings containing S . For a subring $A \subseteq B$, the manuscript says B is generated by S over A when $A \cup S$ generates B and every element of S commutes with every element of A . An element a is a unit if $ax = xa = 1$ for some x ; units form $A^\times$. It is left regular if left multiplication by a is injective, and a left zero divisor otherwise; define right regular and right zero divisor similarly. It is regular if both maps are injective, and a zero divisor if either fails. A division ring has $A^\times = A \setminus \{0\}$. A field is a commutative division ring. An integral domain is a nonzero commutative ring whose nonzero elements are regular.

Example. $\mathbb{Z}$ is a domain but not a field. Every finite domain is a field.

Theorem (Wedderburn). *Every finite division ring is a field.*

10 Modules, ideals, and algebras

10.1 Modules

Definition. A left A -module is an abelian group M with a bilinear map $A \times M \rightarrow M$ satisfying $1m = m$ and $(ab)m = a(bm)$. A module morphism $\varphi : M \rightarrow N$ is additive and satisfies $\varphi(am) = a\varphi(m)$. Write $A\text{-Mod}$ for their category. Right modules are defined analogously, with action $M \times A \rightarrow M$, and form $\text{Mod-}A$.

A left module structure is equivalent to a ring map $A \rightarrow \text{End}_{\text{Ab}}(M)$. In particular $\mathbb{Z}\text{-Mod} = \text{Ab}$ and, for a field k , $k\text{-Mod} = \text{Vect}_k$. Products are formed componentwise; coproducts are the additive direct sums, with $a(x_i) = (ax_i)$ and the canonical inclusions. A submodule $S \leq M$ is an additive subgroup stable under multiplication by elements of A . Examples include $0, M$; images and inverse images of submodules; intersections; and sums $\sum_{\alpha} M_{\alpha}$, consisting of finite sums of elements of the indicated submodules. The submodule generated by $S \subseteq M$ is

$$A(S) = \bigcap_{S \subseteq N \leq M} N = \left\{ \sum_{i \in J} a_i x_i : a_i \in A, x_i \in S, J \text{ finite} \right\}.$$

10.2 Ideals

For subsets of a ring, $S + T = \{s + t\}$ and $ST = \{st\}$ denote setwise operations. A left ideal is an additive subgroup $I \leq A$ with $AI \subseteq I$; a right ideal has $IA \subseteq I$. A two-sided ideal has both properties. In a commutative ring these distinctions disappear. Examples are 0, the unit ideal $(1) = A$, kernels of ring maps, and inverse images of ideals. Intersections and sums of left ideals are left ideals; the same holds for two-sided ideals. The left ideal generated by S is $A(S)$; the two-sided ideal generated by S is

$$A(S)A = \left\{ \sum_{i \in J} a_i x_i b_i : a_i, b_i \in A, x_i \in S, J \text{ finite} \right\}.$$

An ideal is principal if generated by one element. A commutative principal ideal ring has every ideal principal; if it is also a domain, it is a principal ideal domain (PID). For a two-sided ideal I and a submodule $N \leq M$, IN denotes finite sums $\sum_i a_i x_i$ with $a_i \in I$, $x_i \in N$. In particular the product IJ of two ideals is an ideal.

10.3 Quotient modules and rings

For $N \leq M$, the quotient module has action $a(x + N) = ax + N$ and projection $\pi_N : M \rightarrow M/N$. Any module map killing N factors uniquely through π_N . For a two-sided ideal I , the quotient ring A/I has $(a + I)(b + I) = ab + I$. Any ring map killing I factors uniquely through A/I . The ring $\mathbb{Z}/n\mathbb{Z}$ is a domain exactly when n is prime or $n = 0$.

Theorem (Chinese remainder theorem). *If ideals $I_1, \dots, I_n$ satisfy $I_i + I_j = A$ for $i \neq j$, the canonical map gives*

$$A / \bigcap_{k=1}^n I_k \xrightarrow{\sim} \prod_{k=1}^n A / I_k.$$

Proof. The kernel is the indicated intersection, so prove surjectivity. For $n = 2$, choose $z_1 \in I_1$, $z_2 \in I_2$ with $z_1 + z_2 = 1$. Given residues of x_1, x_2 , use $x = z_2x_1 + z_1x_2$. For induction,

$$A = (I_1 + I_2) \cdots (I_1 + I_n) \subseteq I_1 + \prod_{j=2}^n I_j \subseteq I_1 + \bigcap_{j=2}^n I_j.$$

Apply the two-ideal case and induction. □

10.4 Algebras

Let A be commutative. In the broad sense, an A -algebra is an A -module B with an A -bilinear multiplication $B \times B \rightarrow B$. A morphism is an A -linear map preserving multiplication.

Example. A Lie algebra over a field k is such an algebra with bracket $[,]$ satisfying $[x, x] = 0$ and the Jacobi identity

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

An algebra is associative if $x(yz) = (xy)z$, unital if it has an identity, and commutative if $xy = yx$. A unital algebra map preserves 1. **Convention.** From here, “ A -algebra” means unital associative A -algebra, and morphisms preserve 1. Write $\mathbf{Alg}_A$ and $\mathbf{CAlg}_A$ for all and commutative such algebras. Examples: $\mathbb{Z}$ -algebras are rings; A/I is an A -algebra. The algebra A is initial: its unique map to B is $\lambda_B(a) = a1_B$, with image contained in $Z(B)$, since $(a1_B)b = ab = b(a1_B)$.

10.5 Free modules and monoid algebras

The free module on S is $A^{(S)} = \bigoplus_{x \in S} A$, with the map $\delta(x) = \delta_x$. It is universal for set maps $S \rightarrow M$. A module is free with basis $S \subseteq M$ if inclusion induces $A^{(S)} \simeq M$. For commutative A and a monoid M , the monoid algebra $A[M]$ represents monoid maps $M \rightarrow (B, \cdot, 1)$ for A -algebras B .

Proposition. *The monoid algebra exists.*

Proof. Take the free A -module $A^{(M)}$ and define $\delta_x \delta_y = \delta_{xy}$, extending bilinearly:

$$\left(\sum_x a_x \delta_x \right) \left(\sum_y b_y \delta_y \right) = \sum_{x,y} a_x b_y \delta_{xy}.$$

This is associative with identity δ_{1_M} . A monoid map $f : M \rightarrow B$ extends uniquely A -linearly, and the extension is multiplicative on basis elements, hence everywhere. $\square$

As finitely supported functions on M , multiplication is convolution:

$$(f * g)(x) = \sum_{yz=x} f(y)g(z).$$

This follows by collecting the coefficient of δ_x in the product of the basis expansions.

10.6 Group algebras and Dirichlet series

The group algebra $A[G]$ is the monoid algebra with G viewed as a monoid. Its convolution is

$$(f * g)(x) = \sum_{yz=x} f(y)g(z) = \sum_z f(xz^{-1})g(z).$$

For comparison, $L^1(\mathbb{R})$ is an associative nonunital algebra under $(f * g)(x) = \int_{\mathbb{R}} f(x - y)g(y) dy$. A G -module is an abelian group M with an action of G by additive automorphisms. Its morphisms are equivariant additive maps. Restriction along $G \rightarrow \mathbb{Z}[G]$ gives an equivalence $\mathbb{Z}[G]\text{-Mod} \simeq G\text{-Mod}$. A $k[G]$ -module is likewise a k -vector space equipped with a representation $G \rightarrow \text{GL}(V)$. A formal Dirichlet series is $\sum_{n \geq 1} a_n n^{-s}$ with $a_n \in \mathbb{C}$. Termwise addition and multiplication

$$\left(\sum_m \frac{a_m}{m^s} \right) \left(\sum_n \frac{b_n}{n^s} \right) = \sum_k \frac{\sum_{mn=k} a_m b_n}{k^s}$$

make these a commutative $\mathbb{C}$ -algebra. This is the algebra of arithmetic functions $\mathbb{N}_{>0} \rightarrow \mathbb{C}$, with Dirichlet convolution $(a * b)_n = \sum_{d|n} a_d b_{n/d}$.

11 Polynomials and factorization

11.1 Polynomial algebras

The free commutative monoid on S is $\mathbb{N}^{(S)}$. For $S = \{X_1, \dots, X_n\}$, its elements are monomials $X^{\mathbf{d}} = X_1^{d_1} \cdots X_n^{d_n}$, with $\mathbf{d} \in \mathbb{N}^n$ and $|\mathbf{d}| = \sum_i d_i$. The polynomial ring is

$$A[X_1, \dots, X_n] = A[\mathbb{N}^{(S)}] = \bigoplus_{\mathbf{d} \in \mathbb{N}^n} AX^{\mathbf{d}}.$$

The X_i are variables, A is the coefficient ring, and the $X^{\mathbf{d}}$ are standard monomials. A monomial is a scalar multiple of a standard monomial. For nonzero $f = \sum_{\mathbf{d}} a_{\mathbf{d}} X^{\mathbf{d}}$, its total degree is $\max\{|\mathbf{d}| : a_{\mathbf{d}} \neq 0\}$; set $\deg 0 = -\infty$. Degree in X_i treats the other variables as coefficients. The homogeneous component $A[X_1, \dots, X_n]_d$ is spanned by standard monomials of total degree d . Every polynomial decomposes uniquely as $f = \sum_d f^{(d)}$; it is homogeneous of degree d if $f = f^{(d)}$.

Proposition (Universal property). *For a commutative A -algebra B and $\mathbf{b} = (b_1, \dots, b_n) \in B^n$, there is a unique A -algebra map $\varphi_{\mathbf{b}} : A[X_1, \dots, X_n] \rightarrow B$ sending X_i to b_i . It is evaluation:*

$$\varphi_{\mathbf{b}}(f) = f(\mathbf{b}) = \sum_{\mathbf{d}} a_{\mathbf{d}} b_1^{d_1} \cdots b_n^{d_n}.$$

Proof. Combine the universal properties of the monoid algebra and the free commutative monoid to identify the set of maps with B^n . $\square$

For a subring $A \subseteq B$, a tuple $\mathbf{b}$ is algebraically dependent over A if $\ker \varphi_{\mathbf{b}} \neq 0$, and algebraically independent otherwise. A single element c is algebraic or transcendental by the corresponding condition. A tuple is a zero of f if $f(\mathbf{b}) = 0$.

11.2 Prime and maximal ideals

The submodule lattice $\text{Sub}(M)$ is ordered by inclusion; $\text{Sub}_N(M)$ consists of submodules containing N . Inverse image under π_N gives an order-preserving bijection

$$\text{Sub}(M/N) \simeq \text{Sub}_N(M),$$

with inverse $H \mapsto H/N$. For commutative A , $\text{Ideal}(A) = \text{Sub}_A(A)$.

Proposition. *For an ideal $\mathfrak{m}$ of a commutative ring A , the following are equivalent: $\mathfrak{m}$ is maximal among proper ideals; $A/\mathfrak{m} \neq 0$ has only the zero and unit ideals; $A/\mathfrak{m}$ is a field. Such an ideal is called maximal.*

Proof. Use the ideal correspondence, and observe that a nonzero commutative ring has no nontrivial proper ideal exactly when every nonzero element generates the unit ideal. $\square$

Proposition. *For a proper ideal $\mathfrak{p}$, the following are equivalent:*

1. $xy \in \mathfrak{p}$ implies $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$.
2. $IJ \subseteq \mathfrak{p}$ implies $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$ for all ideals I, J .
3. $A/\mathfrak{p}$ is a domain.

Such an ideal is prime.

Proof. If $I \not\subseteq \mathfrak{p}$, choose $x \in I \setminus \mathfrak{p}$; then $xJ \subseteq \mathfrak{p}$ implies $J \subseteq \mathfrak{p}$. Conversely apply (2) to the principal ideals generated by x, y . The quotient characterization is immediate. $\square$

Every maximal ideal is prime. Every proper ideal is contained in a maximal ideal: the union of any chain of proper ideals containing it is again proper, so Zorn's lemma applies. The inverse image of a prime ideal under $\varphi : A \rightarrow B$ is prime, since $A/\varphi^{-1}(\mathfrak{p})$ embeds into the domain $B/\mathfrak{p}$. In $\mathbb{Z}$, $n\mathbb{Z}$ is prime for nonnegative n exactly when $n = 0$ or n is prime. The manuscript calls x a prime element when (x) is a prime ideal.

11.3 Irreducible elements and UFDs

In a domain A , a nonzero nonunit a is irreducible if $a = bc$ implies one of b, c is a unit. Elements a, b are associates if $a = ub$ for a unit u . Associates of irreducibles are irreducible.

Proposition. *A nonzero prime element in a domain is irreducible.*

Proof. If $a = bc$, primality implies, say, $b = ad$. Then $a = adc$; cancellation gives $dc = 1$. $\square$

A factorization of nonzero a into irreducibles is $a = u \prod_{i=1}^n p_i$, with u a unit and each p_i irreducible. A unique factorization domain (UFD), also called factorial, is a domain in which every nonzero element has such a factorization, unique up to reordering and associates.

Proposition. *Every irreducible in a UFD is prime.*

Proof. If $ab = pc \neq 0$, uniqueness forces p to appear in the factorization of a or b . The zero case is immediate in a domain. $\square$

Write $a \mid b$ if $b \in (a)$. A greatest common divisor d of a, b satisfies $c \mid d$ if and only if $c \mid a$ and $c \mid b$. In a UFD, every pair has a gcd, unique up to associates.

11.4 PIDs are UFDs

In a PID, d is a gcd of a, b exactly when $(a, b) = (d)$: divisibility translates to containment of principal ideals. Thus an irreducible p is prime. Indeed, if $ab \in (p)$ and $a \notin (p)$, then $(a, p) = (1)$, whence $b \in b(a, p) = (ba, bp) \subseteq (p)$.

Theorem. *Every PID is a UFD.*

Proof. For uniqueness, a prime factor in one factorization divides some irreducible factor in the other, so they are associates. Cancel and induct. For existence, consider the principal ideals (a) of nonzero nonunits lacking a factorization. A union of an ascending chain is principal, say (c) , and c lies in one term, so the chain stabilizes. Zorn's lemma gives a maximal such ideal (a) . The element a is not irreducible, so $a = bc$ for nonunits b, c . Both (b) and (c) strictly contain (a) , so both elements factor by maximality; then so does a , a contradiction. $\square$

11.5 One-variable polynomials

The leading coefficient of nonzero $f = \sum_d a_d X^d$ is $\text{LC}(f) = a_{\deg f}$. A polynomial is monic if its leading coefficient is 1.

Theorem (Division algorithm). *If $\text{LC}(g)$ is a unit, there are unique $q, r \in A[X]$ with $f = gq + r$ and $\deg r < \deg g$.*

Proof. If $\deg f < \deg g$, take $q = 0, r = f$. Otherwise subtract

$$\text{LC}(f) \text{LC}(g)^{-1} X^{\deg f - \deg g} g$$

from f to lower its degree and induct. For uniqueness, $g(q_1 - q_2) = r_2 - r_1$. Since the leading coefficient of g is a unit, a nonzero left side has degree at least $\deg g$, contrary to the bound on the right. $\square$

Theorem. *If k is a field, $k[X]$ is a PID, hence a UFD.*

Proof. For a nonzero ideal choose a nonzero g of minimal degree. Division of any f in the ideal by g leaves a remainder in the ideal of smaller degree, hence zero. $\square$

The derivative is the A -linear map $D(X^n) = nX^{n-1}$; write $f' = Df$ and $f^{(n)} = D^n f$. It satisfies $(fg)' = f'g + fg'$. For a nonconstant polynomial over a field, characteristic zero implies $f' \neq 0$; in characteristic $p > 0$, $f' = 0$ exactly when $f(X) = g(X^p)$.

11.6 Roots and multiplicities

For a UFD and irreducible p , let $\text{ord}_p(a) = n$ when $a = p^n b$ with $p \nmid b$, and set $\text{ord}_p(0) = \infty$. A nonzero element is a unit times the product of $p^{\text{ord}_p(a)}$ over irreducibles up to associates. For $a \in k$, the multiplicity of a root in f is $\text{ord}_a(f) = \text{ord}_{X-a}(f)$; it is multiple if this is at least 2.

Theorem. *For nonzero $f \in k[X]$:*

1. *There is a unique expansion $f(X) = \sum_{n=0}^{\deg f} b_n(X-a)^n$, with $b_0 = f(a)$ and $b_1 = f'(a)$.*
2. *a is a multiple root exactly when $f(a) = f'(a) = 0$.*
3. *f has at most $\deg f$ roots, counted with multiplicity.*

Proof. Repeatedly divide by $X - a$ for (1); (2) follows. For distinct roots a_i , the product $\prod_i (X - a_i)^{\text{ord}_{a_i}(f)}$ divides f , giving (3). $\square$

A field is algebraically closed if every nonconstant polynomial has a root in it.

Theorem. *Every finite subgroup $U \leq k^\times$ is cyclic.*

Proof. It suffices to show each primary component $U(p)$ is cyclic. Choose $a \in U(p)$ of maximal order p^r . Every element satisfies $x^{p^r} = 1$, so the root bound gives $|U(p)| \leq p^r = |\langle a \rangle|$, proving equality. $\square$

In particular, a finite field has cyclic multiplicative group. An n th root of unity satisfies $\zeta^n = 1$. The manuscript calls it primitive when it generates $k^\times[n]$. Write $\mu(k) = (k^\times)_{\text{tor}}$ for the roots of unity.

12 Localization and polynomial factorization

12.1 Localization

A multiplicative subset S of a commutative ring is a submonoid containing 1. Define $S^{-1}A = A_S$ as pairs (a, s) modulo

$$(a, s) \sim (a', s') \iff \exists t \in S : t(s'a - sa') = 0.$$

Write the class as a/s , and $\ell_S(a) = a/1$. Addition and multiplication are

$$\frac{a}{s} + \frac{a'}{s'} = \frac{as' + a's}{ss'}, \quad \frac{a}{s} \frac{a'}{s'} = \frac{aa'}{ss'}.$$

This is a commutative ring, universal among ring maps from A that send S to units.

Example. 1. For $S = \{1, f, f^2, \dots\}$, write A_f .

2. For prime $\mathfrak{p}$, $S = A \setminus \mathfrak{p}$ gives $A_{\mathfrak{p}}$, with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. A ring with a unique maximal ideal is local.
3. Localizing at all regular elements gives the total ring of fractions $Q(A)$, with $A \hookrightarrow Q(A)$. For a domain this is the fraction field $K(A) = \text{Frac}(A)$.
4. $\mathbb{Z}_{(p)} = \{a/b : a, b \in \mathbb{Z}, p \nmid b\}$ and $\mathbb{Z}_{(0)} = \mathbb{Q}$. The fraction field of $k[X]$ is $k(X) = \{f/g : g \neq 0\}$.

For a UFD and $a = f/g \in K(A)^{\times}$, put $\text{ord}_p(a) = \text{ord}_p(f) - \text{ord}_p(g)$. This is independent of the fraction chosen, since $f/g = f_1/g_1$ implies $fg_1 = f_1g$.

12.2 Polynomials over UFDs

Let A be a UFD and $K = \text{Frac}(A)$. For $f = \sum_n a_n X^n \in K[X]$, put $\text{ord}_p(f) = \min_n \text{ord}_p(a_n)$. For nonzero f , its p -content is $p^{\text{ord}_p(f)}$, and its content $\text{cont}(f)$ is their product, defined up to $A^\times$. A polynomial is primitive if its content is 1 up to units; then it belongs to $A[X]$. Every nonzero $f \in K[X]$ is a scalar times a primitive polynomial.

Lemma (Gauss). $\text{cont}(fg) = \text{cont}(f) \text{cont}(g)$ up to units.

Proof. Reduce to primitive $f, g \in A[X]$. For every irreducible p , neither reduces to zero modulo p . Since $A/(p)$ is a domain, their product does not reduce to zero. Hence fg is primitive. $\square$

Corollary. *A primitive polynomial irreducible in $K[X]$ is irreducible in $A[X]$.*

Proof. If $f = gh$ in $A[X]$, one factor, say g , is a scalar. Then $1 = \text{cont}(f) = g \text{cont}(h)$ up to units, so $g \in A^\times$. $\square$

Theorem. *$A[X]$ is a UFD. Its irreducibles are the irreducibles of A and the primitive polynomials irreducible in $K[X]$.*

Proof. Factor a nonzero polynomial over the UFD $K[X]$, scaling each irreducible factor to be primitive. The remaining scalar is its content, which belongs to A and factors there. For uniqueness, compare factorizations in $K[X]$; primitive factors associated over K are already associated over A , and the remaining contents have unique factorizations in A . $\square$

12.3 Irreducibility criteria

Theorem (Eisenstein). *Let A be a UFD, $K = \text{Frac}(A)$, and $f = \sum_{i=0}^n a_i X^i \in A[X]$ of degree $n \geq 1$. If a prime p satisfies $p \nmid a_n$, $p \mid a_i$ for all $i < n$, and $p^2 \nmid a_0$, then f is irreducible in $K[X]$.*

Proof. Divide out the content and use Gauss's lemma. If $f = gh$ with both factors of positive degree, their leading coefficients are prime to p . Since $\text{ord}_p(a_0) = 1$, assume $g(0)$ is prime to p and $\text{ord}_p(h(0)) = 1$. Let r be the least index with the coefficient of X^r in h prime to p . In the coefficient of X^r in gh , only its product with $g(0)$ is nonzero modulo p , contradicting $p \mid a_r$, since $r < n$. $\square$

Examples: $X^n - a$ is irreducible over $\mathbb{Q}$ for $n \geq 1$ and a square-free integer $a \neq 0, \pm 1$. For prime p , $1 + X + \cdots + X^{p-1}$ is irreducible: replace X by $X + 1$ and apply Eisenstein to $((X + 1)^p - 1)/X$.

Theorem (Reduction criterion). *Let $\varphi : A \rightarrow B$ be a map of domains. If $\deg \varphi(f) = \deg f$ and $\varphi(f)$ is irreducible over $\text{Frac}(B)$, then f has no factorization gh in $A[X]$ with both degrees positive.*

Proof. Such a factorization reduces to $\varphi(g)\varphi(h)$; preservation of the total degree forces both individual degrees to be preserved. $\square$

Example: $X^p - X - 1$ is irreducible over $\mathbb{F}_p$ for prime p , hence over $\mathbb{Q}$.

Proposition (Rational root test). *For $f = a_n X^n + \cdots + a_0 \in A[X]$ over a UFD, if a root is $b/d \in \text{Frac}(A)$ with coprime b, d , then $b \mid a_0$ and $d \mid a_n$.*

12.4 Hilbert basis theorem

A commutative ring is noetherian if every ideal is finitely generated. PIDs are noetherian, and quotients of noetherian rings are noetherian.

Proposition. *A ring is noetherian exactly when every ascending chain of ideals stabilizes.*

Proof. If all ideals are finitely generated, generators of the union of a chain lie in one term, so the chain stabilizes. Conversely, if an ideal were not finitely generated, repeatedly adjoining an element outside the ideal generated so far would produce a strictly ascending chain. $\square$

Theorem (Hilbert). *If A is noetherian, $A[X]$ is noetherian.*

Proof. Suppose an ideal $I \subseteq A[X]$ is not finitely generated. Choose $f_n \in I$ of least degree outside $(f_1, \dots, f_{n-1})$, and write $a_n = \text{LC}(f_n)$. The degrees are nondecreasing. The ideal $(a_1, a_2, \dots)$ is generated by $a_1, \dots, a_N$. Write $a_{N+1} = \sum_{k=1}^N b_k a_k$. Then

$$g = f_{N+1} - \sum_{k=1}^N b_k X^{\deg f_{N+1} - \deg f_k} f_k$$

is outside $(f_1, \dots, f_N)$ but has smaller degree than f_{N+1} , a contradiction. $\square$

Corollary. *Every finitely generated commutative algebra over a noetherian ring is noetherian.*

Proof. It is a quotient of a polynomial ring in finitely many variables. $\square$

13 Field extensions

13.1 Definitions

A subfield $K \subseteq F$ is a subring which is a field. We call F/K an extension. Its degree is $[F : K] = \dim_K F$; it is finite if this dimension is finite. The subfield generated by $S \subseteq F$ over K is $K(S)$, the intersection of subfields containing $K \cup S$. Its elements are rational expressions $f(a_1, \dots, a_n)/g(a_1, \dots, a_n)$ with $a_i \in S$, $f, g \in K[X_1, \dots, X_n]$, and nonzero denominator. Write $K(a_1, \dots, a_n)$ for finite S . The extension is finitely generated if some finite S generates it. For intermediate fields L_1, L_2 of F/K , their compositum is $L_1 L_2 = K(L_1 \cup L_2)$, also called the lift or translation of L_2 to L_1 . For example $\mathbb{Q}(\sqrt{2})\mathbb{Q}(x) = \mathbb{Q}(\sqrt{2})(x)$ in a common overfield. The subring $K[S]$ need not be a field: $\mathbb{Q}[x] \subsetneq \mathbb{Q}(x)$. The extension $\mathbb{Q}(x)/\mathbb{Q}$ is finitely generated as a field but not as a $\mathbb{Q}$ -algebra. A tower is a sequence of nested fields; it is finite if each successive extension is finite.

13.2 Algebraic elements

An element $\alpha \in F$ is algebraic over K if it is a zero of some nonzero polynomial in $K[X]$; otherwise it is transcendental.

Proposition. *For algebraic α , the following conditions on a nonzero polynomial f are equivalent: f is irreducible and $f(\alpha) = 0$; $\ker(K[X] \rightarrow F) = (f)$ for evaluation at α ; f has least degree among nonzero polynomials vanishing at α .*

Proof. The evaluation kernel is a nonzero proper ideal in the PID $K[X]$. Its generator divides every vanishing polynomial. Irreducibility forces an irreducible vanishing polynomial to generate it. A polynomial of least degree cannot factor into two positive-degree factors, since one would also vanish at α . $\square$

The unique monic polynomial satisfying these conditions is the minimal polynomial $\text{Irr}(\alpha, K, X)$.

Proposition. *If this polynomial has degree d , then*

$$K[X]/(\text{Irr}(\alpha, K, X)) \simeq K[\alpha] = K(\alpha), \quad [K(\alpha) : K] = d.$$

Proof. Division yields the basis $1, \alpha, \dots, \alpha^{d-1}$ for $K[\alpha]$. If $g(\alpha) \neq 0$, the minimal polynomial f is coprime to g , so $1 = af + bg$ for some $a, b \in K[X]$. Evaluation gives $1 = b(\alpha)g(\alpha)$. Thus every nonzero element is invertible. $\square$

13.3 Algebraic extensions

Proposition (Tower law). *For $E/F/K$, $[E : K] = [E : F][F : K]$. In particular E/K is finite exactly when both steps are finite.*

Proof. If S is an F -basis of E and T a K -basis of F , the products st form a K -basis of E . □

An extension is algebraic if all its elements are algebraic, and transcendental if it is not algebraic.

Theorem. *An extension is finite if and only if it is finitely generated and algebraic.*

Proof. A finite basis generates the field. For any α in a finite-dimensional extension, sufficiently many powers $1, \alpha, \alpha^2, \dots$ are linearly dependent, giving a polynomial relation. Conversely adjoin finitely many algebraic generators one at a time and apply the tower law and the minimal-polynomial degree formula. □

The elements of F algebraic over K form a subfield. Moreover, an extension is algebraic exactly when it is generated by algebraic elements. A class of extensions is distinguished if E/K belongs to it exactly when both F/K and E/F do, and if membership is preserved by compositum with another base field. Algebraic extensions and finite extensions each form a distinguished class.

13.4 Field embeddings

Let $\mathbf{Field}$ be fields with ring homomorphisms. Every such map is injective, hence is an embedding. For E/K and $\sigma : K \rightarrow F$, an embedding $\tau : E \rightarrow F$ extends σ if $\tau|_K = \sigma$. If F/K is an extension and this restriction is the inclusion, τ is an embedding over K .

$$\begin{array}{ccc} E & \xrightarrow{\tau} & F \\ & \nwarrow \quad \nearrow \sigma & \\ & K & \end{array}$$

If $f \in K[X]$ and $f(\alpha) = 0$, an extension τ sends α to a root of f^σ , obtained by applying σ to the coefficients: apply τ to the equation $\sum a_n \alpha^n = 0$.

Lemma. *A K -embedding $\sigma : E \rightarrow E$ of an algebraic extension is an automorphism.*

Proof. Given $\alpha \in E$, let S be all roots in E of its minimal polynomial and let $E' = K(S)$. Then E'/K is finite, and σ carries E' into itself. An injective linear map on this finite-dimensional space is surjective, so α lies in the image. $\square$

We sometimes write α^σ, E^σ for images. Embeddings respect composita: $\sigma(L_1 L_2) = \sigma(L_1) \sigma(L_2)$.

Proposition. *For a nonconstant $f \in K[X]$, there exist a field E , an embedding $\sigma : K \rightarrow E$, and an element $\alpha \in E$ such that $f^\sigma(\alpha) = 0$.*

Proof. Replace f by an irreducible factor and take the field $K[X]/(f)$. $\square$

13.5 Roots in extension fields

The embedding in the preceding proposition can be realized as an actual inclusion $K \subseteq E$. Choose a set disjoint from K in bijection with the complement of $\sigma(K)$ in the constructed field, extend σ to a bijection, and transport the field operations through it. This produces an extension containing a root of f . Iterating gives an extension where every polynomial in a finite collection has a root. In fact this holds simultaneously for all nonconstant polynomials in $K[X]$.

Proof. Introduce one variable X_f for each nonconstant $f \in K[X]$ and form the ideal $I = (f(X_f) : f)$. It is proper: otherwise an expression $1 = \sum_{i=1}^n g_i f_i(X_{f_i})$ would involve only finitely many variables and polynomials. Choose an extension with roots for that finite collection and evaluate; this would give $1 = 0$. Put I inside a maximal ideal $\mathfrak{m}$. The quotient by $\mathfrak{m}$ is a field where the image of each X_f is a root. Transporting the field structure as above makes it an extension of K . $\square$

13.6 Algebraic closure

In an algebraically closed field every nonconstant polynomial factors into linear factors, by repeatedly extracting a root.

Theorem. *Every field has an algebraically closed extension.*

Proof. Form $K = E_0 \subseteq E_1 \subseteq \cdots$ so that every nonconstant polynomial over E_i has a root in E_{i+1} . In $E = \bigcup_i E_i$, a polynomial has all its finitely many coefficients in one E_i , and therefore has a root in E . $\square$

The elements of E algebraic over K form an algebraically closed algebraic extension $\overline{K}$: an element algebraic over this subfield is algebraic over K and hence belongs to it. For an embedding $\sigma : K \rightarrow L$ and extension E/K , let $\text{Hom}_\sigma(E, L)$ be embeddings $E \rightarrow L$ extending σ .

Proposition. *If $E = K(\alpha)$ is algebraic and L is algebraically closed, $\text{Hom}_\sigma(E, L)$ is nonempty and has as many elements as the distinct roots of $\text{Irr}(\alpha, K, X)^\sigma$ in L .*

Proof. Maps out of $K[X]/(\text{Irr}(\alpha, K, X))$ correspond exactly to choices of the image of X among these roots. $\square$

Theorem. *If E/K is algebraic, L algebraically closed, and $\sigma : K \rightarrow L$, then $\text{Hom}_\sigma(E, L) \neq \emptyset$. If E is also algebraically closed and $L/\sigma(K)$ algebraic, every such embedding is an isomorphism.*

Proof. Order pairs (F, τ) , where $K \subseteq F \subseteq E$ and $\tau : F \rightarrow L$ extends σ , by extension. Unions bound chains, so Zorn gives a maximal pair. If $F \neq E$, adjoin an algebraic element and apply the preceding proposition to extend τ , a contradiction. For the final claim, the image of E is algebraically closed and L is algebraic over it, so equals it. $\square$

Corollary. *Any two algebraically closed algebraic extensions of K are isomorphic over K .*

An *algebraic closure* of K is such an extension $\overline{K}/K$, unique up to isomorphism, which need not be unique.

13.7 Splitting fields

A nonzero polynomial f splits over F if $f = \text{LC}(f) \prod_i (X - \alpha_i)$ in $F[X]$. A splitting field of a family $S \subseteq K[X]$ is an extension where every polynomial in S splits and no smaller intermediate field has this property. It is generated by all the roots, so is algebraic over K .

Theorem. *Any two splitting fields E, F of S are isomorphic over K . If $F \subseteq \overline{K}$, every K -embedding $\sigma : E \rightarrow \overline{K}$ maps E isomorphically onto F .*

Proof. Embed both fields into an algebraic closure. The embedding σ sends the roots of each $f \in S$ to its roots in $\overline{K}$, permuting them with multiplicity. Since F contains exactly all these roots and they generate the splitting fields, $\sigma(E) = F$. $\square$

A splitting field of all polynomials in $K[X]$ is an algebraic closure. Write K_S for a splitting field of S .

14 Normal and separable extensions

14.1 Normality

Theorem. *For $K \subseteq F \subseteq \overline{K}$, the following are equivalent:*

1. *Every K -embedding $F \rightarrow \overline{K}$ induces an automorphism of F .*
2. *F is a splitting field of a family in $K[X]$.*
3. *Every irreducible polynomial in $K[X]$ with a root in F splits over F .*

Proof. Splitting fields give (2) $\Rightarrow$ (1). For (3) $\Rightarrow$ (2), take the minimal polynomials of all elements of F . For (1) $\Rightarrow$ (3), let $\alpha_1 \in F$ be a root of irreducible f . For every other root $\alpha_j \in \overline{K}$, the map $K(\alpha_1) \rightarrow \overline{K}$ sending α_1 to α_j extends to F . By (1) its image lies in F , so $\alpha_j \in F$. $\square$

Such an algebraic extension is normal. Quadratic extensions are normal. The extension $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$ is not normal, since $X^4 - 2$ has nonreal roots outside it. However both steps through $\mathbb{Q}(\sqrt{2})$ are normal, so normal extensions are not a distinguished class. Normality is preserved by base extension (compositum), and if F/K is normal then F/L is normal for intermediate L . If L/K and L'/K are normal, both LL'/K and $(L \cap L')/K$ are normal. These assertions follow from splitting-field descriptions; for the intersection, an irreducible polynomial with a root there splits in both fields.

14.2 Separability and separable degree

A polynomial is separable if it has no multiple roots in an algebraic closure. An algebraic element is separable if its minimal polynomial is; an algebraic extension is separable if all its elements are. For an embedding $\sigma : K \rightarrow L$ into an algebraically closed field, define the separable degree

$$[E : K]_s = |\text{Hom}_\sigma(E, L)|.$$

It does not depend on the chosen algebraically closed target: algebraic closures of the images of K are isomorphic compatibly with K , and every algebraic image lies in such a closure.

Theorem. *For algebraic $E/F/K$,*

$$[E : K]_s = [E : F]_s [F : K]_s.$$

For finite E/K , $[E : K]_s \leq [E : K]$.

Proof. Partition embeddings of E by their restrictions to F ; each fiber has size $[E : F]_s$. For the inequality reduce by a finite generating tower to $E = K(\alpha)$, where the embedding count is the number of distinct roots of the minimal polynomial, at most its degree. $\square$

Thus α is separable exactly when $[K(\alpha) : K]_s = [K(\alpha) : K]$.

14.3 Separable extensions and closure

Theorem. *For a finite extension E/K , the following are equivalent: E/K is separable; it is generated by finitely many separable elements; $[E : K]_s = [E : K]$.*

Proof. The first implies the second. For generators α_i , put $K_i = K(\alpha_1, \dots, \alpha_i)$. Each α_i remains separable over K_{i-1} , so the tower formulas give equality of degrees. Conversely equality for E/K , together with both degree inequalities in the tower through $K(\alpha)$, forces equality for $K(\alpha)/K$ for every $\alpha \in E$. $\square$

An algebraic extension is separable exactly when generated by separable elements: every element lies in a field generated by finitely many of them.

Theorem. *Separable extensions form a distinguished class.*

Proof. Downward passage in towers is immediate. For transitivity, take $\alpha \in F$ in a tower $F/E/K$ of separable extensions, and let S be the finite set of coefficients of its minimal polynomial over E . Both $K(S)/K$ and $K(S, \alpha)/K(S)$ are finite separable; the tower formulas give separability over K . For base extension, separable generators remain separable after enlarging the base, since their new minimal polynomials divide their old ones. $\square$

The separable closure $K^s \subseteq \overline{K}$ consists of elements separable over K . The conjugates of algebraic α are the images $\sigma(\alpha)$ under K -embeddings $K(\alpha) \rightarrow \overline{K}$.

14.4 Primitive element theorem

Let $\text{Sub}_K(E)$ be the intermediate fields of E/K .

Theorem. *For a finite extension E/K , the following are equivalent: $E = K(\alpha)$ for some α ; there are finitely many intermediate fields. A finite separable extension satisfies both conditions. Such an α is a primitive element, and E/K is simple.*

Proof. If K is finite, so is E , whose cyclic multiplicative group has a generator α ; then $E = K(\alpha)$, and finiteness of the intermediate-field set is clear. Assume K infinite. If there are finitely many intermediate fields, choose a maximal simple one $F = K(\alpha)$. For $\beta \in E$, two distinct $c_1, c_2 \in K$ give the same field $K(\alpha + c_1\beta) = K(\alpha + c_2\beta)$. This field contains β and α , so equals $K(\alpha, \beta)$. Maximality gives $\beta \in F$, proving $F = E$. Conversely suppose $E = K(\alpha)$ with minimal polynomial f . To an intermediate field F assign the monic divisor $\text{Irr}(\alpha, F, X)$ of f in $E[X]$. There are finitely many such divisors. The assignment is injective: if the coefficients generate L over K , then $L \subseteq F$ and $[E : L] \leq \deg \text{Irr}(\alpha, F, X) = [E : F]$, forcing $L = F$. Finally, for finite separable E/K , induction reduces to $E = K(\alpha, \beta)$. Let $\sigma_1, \dots, \sigma_n$ be its distinct embeddings into $\overline{K}$, where $n = [E : K]$. The polynomial

$$P(T) = \prod_{i \neq j} (\sigma_i(\alpha) - \sigma_j(\alpha) - T(\sigma_i(\beta) - \sigma_j(\beta)))$$

is nonzero. Choose $c \in K$ with $P(c) \neq 0$. Then $\alpha - c\beta$ has at least n distinct conjugates, so $[K(\alpha - c\beta) : K] \geq n$, and equality of fields follows. $\square$

15 Finite fields and inseparability

15.1 Finite fields

A finite field has prime characteristic p , prime subfield $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, and order p^n , where n is its dimension over $\mathbb{F}_p$. Every automorphism fixes $\mathbb{F}_p$.

Theorem. *For every $n \geq 1$, there is a unique subfield $\mathbb{F}_{p^n} \subseteq \overline{\mathbb{F}_p}$ with p^n elements. It is the splitting field of $X^{p^n} - X$ and consists of its roots. Every finite field of that order is isomorphic to it.*

Proof. The derivative is -1 , so the polynomial has p^n distinct roots in the closure. These roots are closed under multiplication and inverse, and under addition because $(x + y)^{p^n} = x^{p^n} + y^{p^n}$ in characteristic p . They thus form a field. In any field of order p^n , every nonzero element satisfies $x^{p^n-1} = 1$, so all elements are roots. It is therefore a splitting field of the same polynomial, proving uniqueness up to isomorphism and uniqueness inside the closure. $\square$

15.2 Frobenius

For a commutative $\mathbb{F}_p$ -algebra, Frobenius is the endomorphism $\varphi(x) = x^p$.

Theorem. *$\text{Aut}(\mathbb{F}_{p^n})$ is cyclic of order n , generated by Frobenius.*

Proof. Frobenius is injective, hence bijective on the finite field. There are at most n automorphisms by the embedding bound. If d is its order, $\varphi^n = \text{id}$ implies $d \mid n$. Since every element is a root of $X^{p^d} - X$, the root bound gives $p^n \leq p^d$, forcing $d = n$. $\square$

Inside $\overline{\mathbb{F}}_p$, $\mathbb{F}_{p^n} \subseteq \mathbb{F}_{p^m}$ exactly when $n \mid m$. For $q = p^n$, $\mathbb{F}_{q^d}/\mathbb{F}_q$ is normal and separable, and its automorphism group is generated by φ^n . These assertions are left as exercises in the source.

15.3 Inseparable degree

Proposition. *A minimal polynomial over a characteristic-zero field is separable. In characteristic $p > 0$, all roots of a minimal polynomial have the same multiplicity p^μ for some $\mu \geq 0$. For algebraic α ,*

$$[K(\alpha) : K] = p^\mu [K(\alpha) : K]_s,$$

and α^{p^μ} is separable over K .

Proof. Embeddings carrying one root to another extend to the algebraic closure and preserve multiplicities, so these multiplicities agree. An irreducible polynomial with a repeated root has zero derivative. In characteristic zero its derivative cannot vanish. In characteristic p , a zero derivative gives $f(X) = g(X^p)$ with g irreducible. Repeat until $f(X) = h(X^{p^\mu})$ with h separable. The map $x \mapsto x^{p^\mu}$ is injective in a field, and each root of h has exactly one preimage in the algebraic closure, of multiplicity p^μ in f . This proves the degree assertion and separability of α^{p^μ} . $\square$

For finite E/K , $[E : K]_s$ divides $[E : K]$. The inseparable degree is $[E : K]_i = [E : K]/[E : K]_s$, equal to 1 in characteristic zero and a power of p in characteristic p . It equals 1 exactly for separable extensions, and multiplies in towers.

15.4 Purely inseparable extensions

Assume characteristic $p > 0$. An algebraic element α is purely inseparable over K if $\alpha^{p^n} \in K$ for some $n \geq 0$.

Theorem. *For algebraic E/K , the following are equivalent:*

1. $[E : K]_s = 1$.
2. Every $\alpha \in E$ has minimal polynomial $X^{p^n\alpha} - a$ for some $a \in K$.
3. Every element is purely inseparable.
4. The extension is generated by purely inseparable elements.

Such an extension is purely inseparable.

Proof. Having a single embedding is equivalent to the same condition for each simple subextension. Its minimal polynomial then has only one distinct root and degree p^n , hence equals $(X - \alpha)^{p^n} = X^{p^n} - \alpha^{p^n}$. This implies (3) and (4). Conversely any finite set of purely inseparable generators gives a tower of simple extensions, each with only one embedding, so every element in the generated field has separable degree 1. $\square$

Purely inseparable extensions form a distinguished class. If E_0 is the compositum of all subfields of E separable over K , then E_0/K is separable and E/E_0 purely inseparable: each $\alpha \in E$ has a power α^{p^μ} separable over K , hence in E_0 . If L/K is normal, its maximal separable subextension L_0/K is also normal. If E/K is finite separable and F/K finite purely inseparable in a common overfield, then

$$\begin{aligned} [EF : F] &= [E : K]_s = [EF : K]_s, \\ [EF : E] &= [F : K]_i = [EF : K]_i. \end{aligned}$$

Proof. The extension EF/E is purely inseparable and EF/F separable. The separable and inseparable tower formulas, compared with the ordinary tower formula, give the equalities. $\square$

15.5 Frobenius and separability

Write $E^p = \{x^p : x \in E\}$. For finite E/K in characteristic p ,

$$E/K \text{ is separable} \iff E^p K = E.$$

If separable, in fact $E^{p^n} K = E$ for every $n \geq 1$.

Proof. If $E^p K = E$, iteration gives $E^{p^n} K = E$. For finitely many generators choose n large enough that their p^n th powers lie in the maximal separable subfield E_0 ; then $E \subseteq E_0$. Conversely $E/E^p K$ is purely inseparable, and is also separable if E/K is; it must be trivial. $\square$

15.6 Perfect fields

For normal L/K in characteristic p , let $G = \text{Aut}_K(L)$ and $L^G = \{x : \sigma(x) = x \text{ for all } \sigma \in G\}$.

Proposition. *L^G/K is purely inseparable, L/L^G is separable, and, if L_0 is the maximal separable subextension of L in L , then $L = L^G L_0$ and $L^G \cap L_0 = K$.*

Proof. An embedding of $K(\alpha)$ for $\alpha \in L^G$ extends to L and, by normality, to an automorphism there. It must fix α , so α has only one conjugate and is purely inseparable. For separability, the product over the distinct G -conjugates of any $\alpha \in L$ is separable and has coefficients in L^G ; one may reduce to the finite normal splitting field of its minimal polynomial. The intersection is both separable and purely inseparable over K , so equals K . The extension $L/L^G L_0$ is likewise both types and hence trivial. $\square$

A field is perfect if it has characteristic zero, or characteristic p and $K^p = K$.

Corollary. *Every algebraic extension of a perfect field is separable and is itself perfect.*

Proof. Reduce separability to a finite normal extension L/K . Its fixed field L^G is finite purely inseparable, so for some n , $K^{p^n} \subseteq (L^G)^{p^n} \subseteq K$. Perfectness gives $(L^G)^{p^n} = K$, whence $[L^G : K] = 1$. Thus L/K is separable. For a finite subextension E/K , the preceding criterion gives $E = E^p K = E^p K^p = E^p$. The general algebraic case is the union of its finite subextensions. $\square$

16 Galois theory

16.1 Galois extensions and correspondence

For a subgroup $G \leq \text{Aut}(L)$, its fixed field is $L^G = \{x : \sigma(x) = x \ \forall \sigma \in G\}$. An algebraic extension L/K is Galois if normal and separable; its Galois group is $\text{Gal}(L/K) = \text{Aut}_K(L)$.

Theorem (Fundamental theorem, finite case). *For finite Galois L/K and $G = \text{Gal}(L/K)$, the maps*

$$H \longmapsto L^H, \quad F \longmapsto \text{Gal}(L/F)$$

are inverse order-reversing bijections between subgroups and intermediate fields. A subgroup H is normal exactly when L^H/K is normal, in which case restriction gives $G/H \simeq \text{Gal}(L^H/K)$.

We establish the ingredients below. For any Galois extension, possibly infinite, $L^G = K$. Every intermediate extension L/F is Galois, and $F \mapsto \text{Gal}(L/F)$ is injective, since its fixed field is F . If $H_i = \text{Gal}(L/F_i)$, then

$$H_1 \cap H_2 = \text{Gal}(L/F_1 F_2), \quad L^{\langle H_1, H_2 \rangle} = F_1 \cap F_2,$$

and $F_1 \subseteq F_2$ exactly when $H_1 \supseteq H_2$. These follow directly by testing which elements are fixed. A finite separable extension E/K lies in a finite Galois extension L/K , its normal closure (the splitting field of minimal polynomials of generators). Thus it has only finitely many intermediate fields, also following from the primitive element theorem.

16.2 Artin's theorem

For a separable algebraic extension, the degrees of simple subextensions are bounded by a finite integer exactly when the whole extension is finite; in that case their maximum is $[E : K]$. Indeed, choose a simple subextension of maximal degree, and use the primitive element theorem on the field it generates with any other element.

Theorem (Artin). *If a finite group G acts on a field L by automorphisms and $K = L^G$, then L/K is finite Galois, $\text{Gal}(L/K) = G$, and $[L : K] = |G|$.*

Proof. For $\alpha \in L$, take the product $f(X) = \prod_{\beta \in G\alpha} (X - \beta)$. The action permutes its factors, so $f \in K[X]$. It is separable, splits over L , has α as a root, and has degree at most $|G|$. Thus L/K is separable and normal, and the finite bound on simple degrees gives $[L : K] \leq |G|$. Since $G \leq \text{Gal}(L/K)$ and the latter has size $[L : K]$, equality follows. $\square$

For finite Galois L/K , apply Artin to each $H \leq G$ to get $H = \text{Gal}(L/L^H)$, proving surjectivity in the correspondence. For any Galois L/K and intermediate F , put $H = \text{Gal}(L/F)$. Then F/K is normal exactly when $H \trianglelefteq G$, since $\sigma(F) = F$ exactly when $\sigma H \sigma^{-1} = H$. If normal, restriction is surjective onto $\text{Gal}(F/K)$ (extend embeddings to L), with kernel H , so $G/H \simeq \text{Gal}(F/K)$.

16.3 Composita and products

Theorem. *If L/K is Galois and F/K is in a common overfield, then LF/F and $L/(L \cap F)$ are Galois, and restriction gives*

$$\mathrm{Gal}(LF/F) \xrightarrow{\sim} \mathrm{Gal}(L/(L \cap F)).$$

Proof. Normality and separability are preserved by base extension. Restriction is injective since L and F generate LF . Its image fixes exactly $L \cap F$. In the finite case the Galois correspondence proves surjectivity; in the infinite case the image is closed in the profinite topology, so the same conclusion follows from the infinite correspondence. $\square$

For finite Galois L/K , this implies $[LF : F] \mid [L : K]$. If L_1, L_2 are Galois over K , so is $L_1 L_2$, and restriction gives an injection

$$\mathrm{Gal}(L_1 L_2 / K) \hookrightarrow \mathrm{Gal}(L_1 / K) \times \mathrm{Gal}(L_2 / K).$$

It is an isomorphism if $L_1 \cap L_2 = K$: the subgroups fixing each field map onto the two factors separately by the preceding theorem. More generally, if $L_i \cap \prod_{j \neq i} L_j = K$ for all i , the Galois group of the compositum is the product of the individual Galois groups. Conversely, if finite Galois L/K has group $G_1 \times \cdots \times G_n$, let $L_i = L^{\ker \pi_i}$. Then $\mathrm{Gal}(L_i / K) \simeq G_i$, $L = L_1 \cdots L_n$, and $L_i \cap \prod_{j \neq i} L_j = K$. A Galois extension is abelian or cyclic if its group is. Composita and base extensions of abelian extensions are abelian. Subextensions and upper steps of abelian (respectively cyclic) extensions are abelian (respectively cyclic).

16.4 Quadratic and cubic examples

The Galois group of a separable polynomial is the Galois group of its splitting field. Its faithful permutation action on the roots embeds it in S_n , where n is the degree. If $\text{char } K \neq 2$ and $a \in K^\times$ is not a square, $X^2 - a$ is irreducible and separable. Its splitting field is $K(\alpha)$, $\alpha^2 = a$, and the nontrivial automorphism sends α to $-\alpha$. Every quadratic extension has this form: for $\beta^2 + b\beta + c = 0$, replace β by $\alpha = \beta + b/2$, giving $\alpha^2 = b^2/4 - c$. Assume $\text{char } K \neq 2, 3$. By translating the variable, a monic cubic becomes $f = X^3 + aX + b$. If it has no root in K , it is irreducible. For a root α , $[K(\alpha) : K] = 3$, and the splitting-field group is A_3 or S_3 . Let its roots be $\alpha_1, \alpha_2, \alpha_3$, and set

$$\delta = (\alpha_1 - \alpha_2)(\alpha_2 - \alpha_3)(\alpha_1 - \alpha_3), \quad \Delta = \delta^2.$$

Every automorphism sends δ to its sign times δ , so $\Delta \in K$. Direct computation gives $\Delta = -4a^3 - 27b^2$. Consequently

$$G = A_3 \iff \delta \in K \iff \Delta \in (K^\times)^2.$$

For $X^3 - 2$ over $\mathbb{Q}$, $\Delta = -27 \cdot 2^2$ is not a square, giving S_3 and a nonnormal real cubic field. For $X^3 - 3X + 1$, the rational root test gives irreducibility and $\Delta = 81$, so the group is A_3 .

16.5 A quartic and the generic polynomial

For $f = X^4 - 2$, Eisenstein gives irreducibility. Let $\alpha = \sqrt[4]{2} > 0$. Its roots are $\pm\alpha, \pm i\alpha$, so the splitting field is $L = \mathbb{Q}(\alpha, i)$ and $[L : \mathbb{Q}] = 8$, since $i \notin \mathbb{Q}(\alpha) \subseteq \mathbb{R}$. Let τ fix α and send i to $-i$, and let σ fix i and send α to $i\alpha$. Then $\sigma^4 = \tau^2 = 1$ and $\tau\sigma = \sigma^{-1}\tau$. Thus

$$\text{Gal}(L/\mathbb{Q}) = \langle \sigma, \tau \mid \sigma^4, \tau^2, \tau\sigma\tau\sigma \rangle \simeq D_8,$$

the symmetry group of a square. The manuscript's subgroup and field lattices are recorded by this correspondence; inclusion is reversed.

Subgroup H	Fixed field L^H
G	$\mathbb{Q}$
$\langle \sigma^2, \tau \rangle$	$\mathbb{Q}(\alpha^2)$
$\langle \sigma \rangle$	$\mathbb{Q}(i)$
$\langle \sigma^2, \sigma\tau \rangle$	$\mathbb{Q}(i\alpha^2)$
$\langle \tau \rangle$	$\mathbb{Q}(\alpha)$
$\langle \sigma^2\tau \rangle$	$\mathbb{Q}(i\alpha)$
$\langle \sigma^2 \rangle$	$\mathbb{Q}(\alpha^2, i)$
$\langle \sigma\tau \rangle$	$\mathbb{Q}((1+i)\alpha)$
$\langle \sigma^3\tau \rangle$	$\mathbb{Q}((1-i)\alpha)$
$\{1\}$	L

For algebraically independent $t_1, \dots, t_n$ over a field k , S_n acts on $L = k(t_1, \dots, t_n)$ by permuting them. The polynomial

$$\prod_i (X - t_i) = X^n - s_1 X^{n-1} + s_2 X^{n-2} - \dots + (-1)^n s_n$$

has elementary symmetric coefficients $s_j = \sum_{|I|=j} \prod_{i \in I} t_i$. Its fixed field is $L^{S_n} = k(s_1, \dots, s_n)$: inclusion is clear and the splitting-field degree is at most $n!$, while Artin gives $[L : L^{S_n}] = n!$. Thus S_n occurs as a Galois group.

16.6 Algebraic closure of the complex numbers

Theorem. $\mathbb{C} = \mathbb{R}(i)$ is algebraically closed.

Proof. Use that $\mathbb{R}$ is ordered, every positive real is a square, and every odd-degree real polynomial has a real root. First every $a + bi \in \mathbb{C}$ has a square root $c + di$, where

$$c^2 = \frac{a + \sqrt{a^2 + b^2}}{2}, \quad d^2 = \frac{-a + \sqrt{a^2 + b^2}}{2},$$

with signs chosen so $2cd = b$. Second, if $L/\mathbb{R}$ is finite Galois and H is a Sylow 2-subgroup, its fixed field has odd degree. A primitive element has an odd-degree irreducible polynomial, hence a real root, forcing the fixed field to be $\mathbb{R}$. Thus the Galois group is a 2-group. Third, if $\mathbb{C} \subseteq L$ in such an extension and $H = \text{Gal}(L/\mathbb{C})$ were nontrivial, it would have an index-2 subgroup. The fixed field would be quadratic over $\mathbb{C}$, impossible since every complex number has a square root. Hence $L = \mathbb{C}$. Any algebraic element over $\mathbb{C}$ lies in a finite normal extension of $\mathbb{R}$ containing i , and therefore already lies in $\mathbb{C}$. $\square$

Proposition. If an irreducible $f \in \mathbb{Q}[X]$ has prime degree p and exactly two nonreal roots, its Galois group is S_p .

Proof. The transitive group has order divisible by p , so contains a p -cycle by Cauchy. Complex conjugation is a transposition. A p -cycle and any transposition generate S_p : conjugating the transposition produces edges of a connected graph on the p letters, whose edge transpositions generate the symmetric group. $\square$

Example: $X^5 - 4X + 2$ is Eisenstein at 2 and has exactly three real roots; its group is S_5 .

16.7 Reduction modulo primes

Proposition. *Let $f \in \mathbb{Z}[X]$ be monic and suppose that its reduction $\bar{f} \in \mathbb{F}_p[X]$ is separable. Choose a prime of the splitting field $\mathbb{Q}_f$ above p . Reduction at this prime identifies the roots of f with those of $\bar{f}$ and identifies the corresponding decomposition group with $\text{Gal}((\mathbb{F}_p)_{\bar{f}}/\mathbb{F}_p)$. In particular, after this choice, there is an embedding*

$$\text{Gal}((\mathbb{F}_p)_{\bar{f}}/\mathbb{F}_p) \hookrightarrow \text{Gal}(\mathbb{Q}_f/\mathbb{Q})$$

compatible with the two root actions.

The proof uses decomposition groups and reduction at an unramified prime and is deferred to a later course. For $f = X^5 - X - 1$, reduction modulo 5 is irreducible, and modulo 2,

$$\bar{f} = (X^2 + X + 1)(X^3 + X^2 + 1).$$

Thus the Galois group contains a 5-cycle and a disjoint product of a 2-cycle and a 3-cycle. Cubing the latter gives a transposition, hence the group is S_5 .

17 Roots of unity and cyclotomic fields

17.1 Roots of unity

For $\text{char } K \nmid n$, the group $\mu_n(\overline{K})$ is cyclic of order n . If ζ is a generator, $K(\zeta)/K$ is Galois and

$$\text{Gal}(K(\zeta)/K) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times, \quad \sigma(\zeta) = \zeta^{i(\sigma)}.$$

Indeed $X^n - 1$ is separable, embeddings send a primitive root to another primitive root, and composition multiplies exponents. Consequently the extension is abelian of degree dividing Euler's totient $\phi(n)$.

Theorem. *Over $\mathbb{Q}$, this map is an isomorphism.*

Proof. Let f be the minimal polynomial of a primitive ζ_n and write $X^n - 1 = fh$. Gauss's lemma gives $f, h \in \mathbb{Z}[X]$. It suffices to show ζ_n^p is a root of f for every prime $p \nmid n$. Otherwise it is a root of h , so $f(X) \mid h(X^p)$ in $\mathbb{Z}[X]$. Modulo p , $h(X^p) = h(X)^p$, so $\bar{f}$ and $\bar{h}$ have a common factor. Then $X^n - 1$ is not separable modulo p , a contradiction. $\square$

For coprime m, n , $\mathbb{Q}(\zeta_m)\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{mn})$ and $\mathbb{Q}(\zeta_m) \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}$. This follows from the degree formula and $(\mathbb{Z}/mn\mathbb{Z})^\times \simeq (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$.

17.2 Cyclotomic polynomials

The n th cyclotomic polynomial is

$$\Phi_n(X) = \prod_{a \in (\mathbb{Z}/n\mathbb{Z})^\times} (X - \zeta_n^a) = \text{Irr}(\zeta_n, \mathbb{Q}, X) \in \mathbb{Z}[X].$$

Useful identities (exercises):

$$X^n - 1 = \prod_{d|n} \Phi_d(X), \quad \Phi_p(X) = 1 + X + \cdots + X^{p-1},$$

$$\Phi_{p^r}(X) = \Phi_p(X^{p^{r-1}}),$$

$$\Phi_n(X) = \Phi_{p_1 \cdots p_s}(X^{p_1^{r_1-1} \cdots p_s^{r_s-1}}) \quad (n = \prod_i p_i^{r_i}),$$

$$\Phi_{2n}(X) = \Phi_n(-X) \quad (n > 1 \text{ odd}),$$

$$\Phi_{pn}(X) = \frac{\Phi_n(X^p)}{\Phi_n(X)} \quad (p \nmid n),$$

$$\Phi_n(X) = \prod_{d|n} (X^{n/d} - 1)^{\mu(d)}.$$

Here $\mu(n) = 0$ if n is not square-free; $\mu(n) = (-1)^r$ if n is a product of r distinct primes, including $\mu(1) = 1$.

17.3 Quadratic fields and cyclotomic extensions

For odd prime p , the quadratic residue symbol $\chi(v) = \left(\frac{v}{p}\right) : \mathbb{F}_p^\times \rightarrow \{\pm 1\}$ is the quotient by the subgroup of squares. Thus $\chi(v) = 1$ exactly when v is a nonzero square.

Theorem. *For primitive p th root ζ and $\delta = \sum_{v \in \mathbb{F}_p^\times} \chi(v) \zeta^v$, one has $\delta^2 = \chi(-1)p$. Every quadratic extension of $\mathbb{Q}$ is contained in a cyclotomic extension.*

Proof. In the square of the Gauss sum, substitute $\nu = \mu v$ to obtain

$$\begin{aligned} \delta^2 &= \sum_{v \neq 0} \chi(v) \sum_{\mu \neq 0} \zeta^{\mu(v+1)} \\ &= \chi(-1)(p-1) - \sum_{v \neq 0, -1} \chi(v) \\ &= p\chi(-1), \end{aligned}$$

since $\sum_{v \neq 0} \chi(v) = 0$. Thus $\mathbb{Q}(\sqrt{\pm p})$ is cyclotomic for the appropriate sign. Adjoining i supplies the other sign; $(1+i)^2 = 2i$ supplies $\sqrt{2}$ inside a cyclotomic field. Products of these square roots cover every quadratic field. $\square$

Theorem (Kronecker–Weber). *Every finite abelian extension of $\mathbb{Q}$ is contained in a cyclotomic extension.*

18 Characters, norm, and trace

18.1 Linear independence of characters

A one-dimensional character of a monoid G over a field K is a monoid homomorphism $\chi : G \rightarrow K^\times$. It is trivial if constantly 1. Linear independence is taken in the K -vector space of all functions $G \rightarrow K$.

Theorem (Artin). *Distinct characters $\chi_1, \dots, \chi_n$ are linearly independent.*

Proof. Suppose $\sum_i a_i \chi_i = 0$ is a nontrivial relation of least length, with all $a_i \neq 0$. Then $n \geq 2$. Choose z with $\chi_1(z) \neq \chi_2(z)$. Evaluating at zx gives $\sum_i a_i \chi_i(z) \chi_i(x) = 0$. Subtract this relation divided by $\chi_1(z)$ from the original to obtain

$$\sum_{i=2}^n a_i \left(1 - \frac{\chi_i(z)}{\chi_1(z)}\right) \chi_i = 0.$$

Its coefficient for $i = 2$ is nonzero, contradicting minimality. $\square$

In particular, distinct field automorphisms of a finite normal extension are linearly independent over the extension field, by viewing them as characters on its multiplicative group.

18.2 Norm and trace

For finite E/K and $\alpha \in E$, let $m_\alpha : E \rightarrow E$ be multiplication by α , a K -linear map. Define

$$N_{E/K}(\alpha) = \det(m_\alpha), \quad \text{Tr}_{E/K}(\alpha) = \text{Tr}(m_\alpha).$$

The norm is a homomorphism $E^\times \rightarrow K^\times$, and trace is K -linear. If $\text{Irr}(\alpha, K, X) = X^d + a_1 X^{d-1} + \cdots + a_d$, the matrix of multiplication by α in $1, \alpha, \dots, \alpha^{d-1}$ is the companion matrix. Hence

$$N_{K(\alpha)/K}(\alpha) = (-1)^d a_d, \quad \text{Tr}_{K(\alpha)/K}(\alpha) = -a_1.$$

If $\sigma_1, \dots, \sigma_r : E \rightarrow \overline{K}$ are the distinct embeddings over K and $q = [E : K]_i$, then

$$N_{E/K}(\alpha) = \left(\prod_{i=1}^r \sigma_i(\alpha) \right)^q, \quad \text{Tr}_{E/K}(\alpha) = q \sum_{i=1}^r \sigma_i(\alpha).$$

Proof. For $E = K(\alpha)$, factor the minimal polynomial as $\prod_i (X - \sigma_i \alpha)^q$ and compare coefficients. In general, a basis over $K(\alpha)$ gives repeated blocks for multiplication by α , so norm is raised to $[E : K(\alpha)]$ and trace multiplied by it. Each embedding of $K(\alpha)$ has $[E : K(\alpha)]_s$ extensions to E ; combine with multiplicativity of the inseparable degree. $\square$

18.3 Properties of norm and trace

Theorem (Transitivity). *For finite $F/E/K$,*

$$N_{F/K} = N_{E/K} \circ N_{F/E}, \quad \text{Tr}_{F/K} = \text{Tr}_{E/K} \circ \text{Tr}_{F/E}.$$

Proof. Partition embeddings of F into an algebraic closure by their restrictions to E , and apply the product and sum formulas, with inseparable degrees multiplying in the tower. $\square$

Theorem. *If E/K is finite separable, its trace is nonzero and the bilinear pairing $(x, y) \mapsto \text{Tr}_{E/K}(xy)$ induces an isomorphism $E \xrightarrow{\sim} \text{Hom}_K(E, K)$.*

Proof. The trace is the sum of distinct embeddings, which is nonzero by linear independence of characters. If $x \neq 0$, multiplication by x is bijective, so $y \mapsto \text{Tr}(xy)$ is nonzero. Thus the induced linear map is injective; source and target have equal finite dimension, so it is an isomorphism. $\square$

Every basis $\omega_1, \dots, \omega_n$ therefore has a unique dual basis $\omega'_1, \dots, \omega'_n$ with $\text{Tr}(\omega_i \omega'_j) = \delta_{ij}$. Lang gives an explicit dual basis for $1, \alpha, \dots, \alpha^{d-1}$.

19 Cyclic extensions and radicals

19.1 Hilbert's theorem 90

Theorem (Hilbert 90). *Let L/K be finite cyclic with $\text{Gal}(L/K) = \langle \sigma \rangle$ of order n . Then*

$$\{\beta \in L^\times : N_{L/K}(\beta) = 1\} = \{\alpha/\sigma(\alpha) : \alpha \in L^\times\}.$$

Proof. Norms of the displayed quotients are 1. Conversely let $N(\beta) = 1$ and define the K -linear operator $T(x) = \beta\sigma(x)$. Then $T^n = \text{id}$. Linear independence of the distinct automorphisms implies $1 + T + \cdots + T^{n-1}$ is a nonzero operator. Choose θ such that $\alpha = \sum_{i=0}^{n-1} T^i(\theta) \neq 0$. Since $(1 - T)\alpha = (1 - T^n)\theta = 0$, we have $\alpha = \beta\sigma(\alpha)$. $\square$

19.2 Cyclic extensions by radicals

Theorem. *Suppose $\text{char } K \nmid n$ and K contains all n th roots of unity.*

1. *A cyclic extension L/K of degree n has $L = K(\alpha)$ and minimal polynomial $X^n - a$ for some $a \in K$.*
2. *If α is a root of $X^n - a$, then $K(\alpha)/K$ is cyclic. Its degree d divides n , and $\alpha^d \in K$.*

Proof of (1). Let $\zeta \in K$ be primitive. Its norm from L is $\zeta^n = 1$. Hilbert 90, applied to ζ^{-1} , gives $\alpha \neq 0$ with $\sigma(\alpha) = \zeta\alpha$. The n distinct images $\zeta^i\alpha$ show $[K(\alpha) : K] = n$, so $L = K(\alpha)$. Also α^n is fixed by σ , hence lies in K , and the minimal polynomial is $X^n - \alpha^n$. $\square$

Proof of (2). If $a = 0$, the extension is trivial. Otherwise $K(\alpha)$ contains every root $\zeta\alpha$ of $X^n - a$, so it is a separable splitting field. Every automorphism satisfies $\sigma(\alpha) = \omega(\sigma)\alpha$ for $\omega(\sigma) \in \mu_n(K)$. The map ω is an injective homomorphism, so the group is cyclic of order $d \mid n$. For a generator, $\omega(\sigma)^d = 1$, whence α^d is fixed and belongs to K . $\square$

19.3 Solvable extensions

A Galois extension is solvable if its Galois group is. A finite separable extension E/K is solvable if it is contained in a finite solvable Galois extension L/K .

Proposition. *Solvable finite extensions form a distinguished class.*

Proof. For base extension, take a solvable Galois extension E'/K containing E . The Galois group of $E'F/F$ embeds in that of E'/K , so EF/F is solvable. Passage down a tower follows from subgroups and quotients of solvable groups being solvable. For transitivity, suppose $K \subseteq F \subseteq E$ with both steps solvable. Choose finite solvable Galois F'/K containing F , and finite solvable Galois L/F' containing EF' . Take the compositum M of all K -conjugates of L . Each conjugate is solvable Galois over F' ; consequently $\text{Gal}(M/F')$ embeds in a finite product of solvable groups and is solvable. The exact sequence

$$1 \rightarrow \text{Gal}(M/F') \rightarrow \text{Gal}(M/K) \rightarrow \text{Gal}(F'/K) \rightarrow 1$$

shows M/K solvable, and $E \subseteq M$. □

19.4 Radical extensions and Abel–Ruffini

In characteristic zero, a finite extension F/K is solvable by radicals if it is contained in a finite extension E/K admitting a tower

$$K = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_m = E,$$

where $E_{i+1} = E_i(\alpha_i)$ and $\alpha_i^{n_i} \in E_i$ for some positive integer n_i .

Theorem. *In characteristic zero, a finite extension is solvable if and only if it is solvable by radicals.*

Proof. Suppose $E \subseteq L$ with L/K finite solvable Galois. Let m be the product of primes dividing $[L : K]$, and put $F = K(\mu_m)$. This is abelian over K and is itself obtained by adjoining roots of unity. The solvable group $\text{Gal}(LF/F)$ has a cyclic refinement with prime-order factors. By Galois correspondence there is a tower from F to LF with cyclic prime-degree steps. Each base contains the required roots of unity, so each step adjoins a root of $X^p - a$. Thus LF/K , and hence E/K , is solvable by radicals. Conversely, consider a radical tower containing E . Adjoin to K all roots of unity required by its finitely many exponents, forming an abelian extension F/K . After base extension to F , each radical step is cyclic by the preceding theorem. Hence the resulting extension over F is solvable by the distinguished-class property. Since F/K is solvable, transitivity and passage to subextensions show E/K solvable. $\square$

Corollary (Abel–Ruffini). *Let $a_1, \dots, a_d$ be algebraically independent over $\mathbb{Q}$, put $K = \mathbb{Q}(a_1, \dots, a_d)$, and let*

$$f(X) = X^d + a_1 X^{d-1} + \cdots + a_d.$$

For $d \geq 5$, its splitting field K_f/K is not solvable by radicals.

Proof sketch. The generic polynomial has Galois group S_d , which is not solvable for $d \geq 5$. $\square$